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REVIEW

Research advancements in machine learning-assisted design of reinforced composite radiation shields

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Abstract
This systematic review explores research advancements in the field of machine learning-assisted design of reinforced composite radiation shields. The goal is to break down ongoing progressions and accomplishments in optimizing shielding materials through the integration of machine learning (ML) strategies. A precise search was directed across conspicuous data bases, including PubMed, IEEE Xplore, Google Scholar, and Scopus, from 2020 to 2024. Consideration models enveloped examinations zeroed in on the utilization of ML in the design and optimization of built-up composite radiation shields. A total of 194 studies meeting the eligibility criteria were included for detailed analysis. Study selection was based on predefined criteria, considering relevance to the integration of ML in the design of composite shields. Data extraction involved key parameters such as material composition, geometric design, and processing techniques used in the optimization process. Research advancements identified in the ML-assisted design of reinforced composite radiation shields include precise tailoring of material composition for enhanced radiation absorption, geometric optimization strategies for improved shielding efficacy, and successful applications across diverse industries such as healthcare, aerospace, and nuclear technology. This systematic review highlights the transformative impact of ML on the design of reinforced composite radiation shields. Recent achievements indicate a clear understanding of material interactions, leading to the creation of highly efficient shielding materials. The integration of computational tools and experimental validation has opened new frontiers in safety, efficiency, and innovation. Future research directions should focus on interpretability, addressing challenges in data availability, and ensuring ethical considerations in the design process.

Keywords Machine learning · Reinforced composite · Shields · Radiation · Protection

1 Introduction

The usage of radiation has been consistently developing throughout the past a very long time in different fields including clinical, modern, and horticultural fields. In a period characterized by mechanical progression, the union of state-of-the-art advances has added to this consistent development across different areas. One such domain that has witnessed remarkable progress according to Malekar et al., (2023) is the field of radiation shielding, which is a critical facet of medical imaging. The performance of a material as a radiation shield is usually assessed by its capacity to halt the penetration of the incident radiation through different interaction mechanisms. A desirable shielding material is capable of attenuating gamma radiation with minimal alteration to its mechanical, thermal and electrical properties as well as its chemical and physical stability (Abu-Saleem et al. 2021). The design and optimization of radiation shields, particularly those composed of reinforced composites, represent a pivotal challenge in shielding both human health and the integrity of critical equipment in radiation-intensive environments.

Radiation shielding is significant in various applications across enterprises, including medication, nuclear power, aviation, and exploration. Shetgaonkar and Kumar (2022) maintained that radiation protection shields individuals, hardware, and the environment from ionizing radiation's unsafe impacts. Examples include shielding patients and healthcare workers during medical procedures (More et al. 2021), containing radiation in nuclear facilities (Malekzadeh

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et al. 2020), shielding aircraft (Turczyn et al. 2020).and spacecraft from cosmic radiation (Naito et al. 2021), and ensuring safety in radiography (Yunus et al. 2023) and non-destructive testing (Cho et al. 2020). Shielding is also vital in military defense (Lee et al. 2021), environmental protection (El-Khatib et al. 2020), and safety screening. Proper shielding design and materials are strategic to minimize radiation exposure and associated health risks in these diverse applications.

Reinforced composite materials are valuable in radiation shielding due to their versatility and effectiveness in shielding against ionizing radiation. These composites, often reinforced with heavy metals such as lead (Alresheedi et al. 2023), tungsten (Abulyazied et al. 2023), bismuth (Özdemir et al. 2023), or biomaterials such as biodegradable polymers (Nath et al. 2020), and natural fibers (Hemath et al. 2020) offer high-density shielding while remaining lightweight. They can be customized for various applications, resist corrosion, and provide structural integrity. Composite materials are also tunable, reducing toxicity concerns, and offering thermal and electrical insulation properties (Ihueze et al. 2023). Overall, reinforced composites are a versatile and innovative solution for radiation shielding, addressing specific needs across industries while offering advantages over traditional materials like lead.

Historically, the design process for reinforced composite based radiation shields has been a complex, labor-intensive process, often constrained by one factor at a time methodologies (Okafor et al. 2021). These traditional methods for radiation shield design always only focus on either the structure or the components of the shields rather than at the same time, which largely affects the shielding performance of the facilities. However, in recent years, the marriage of ML and materials science has revolutionized the approach to reinforced composite material-based radiation shielding design (Okafor et al. 2023; Shi et al. 2022). Bilmez et al., (2022) proposed a new ML based approach is proposed to model gamma-ray shielding behavior of composites alternative to theoretical calculations. This synergy has obviously accelerated the development of efficient, lightweight, and highly protective radiation shields.

ML is integrated into material design to accelerate the discovery and optimization of engineering materials. It provides a new means of screening novel materials with good performance, developing quantitative structure–activity relationships (Guo et al. 2021). This process involves generating and collecting extensive material data, cleaning, and transforming data into a format that is more easily used to build a predictive mode, extracting meaningful features, and training ML models. ML shows good applicability in classification, regression and other tasks related to high-dimensional data and it is capable of building systems with the ability to continuously improve their performance on a given task based on the experience obtained from the data they are exposed to. They reduce experimental costs and time because of their capability to learn from previous computations which produce reliable, repeatable choices and results and thus has enabled faster innovation. ML in material design has vast potential to transform industries reliant on advanced shielding materials, such as bismuth-based composites, tungsten-based composites, polymer-metal hybrid composites, graphene, and carbon nanotube composites.

The systematic review of literature is motivated by the burgeoning intersection of ML and materials science, particularly in the field of radiation shielding. With the escalating demand for efficient and innovative radiation protection solutions, integrating ML into the design process of reinforced composite shields holds great promise. Traditional methods radiation shielding may fall short in optimizing complex composite materials, necessitating a comprehensive examination of the recent advancements facilitated by ML. The motivation also stems from identified gaps in the existing research. The growing need for improved radiation protection solutions which combine efficiency with low cost and flexibility has led to a necessity for discovering new shielding approaches. Advances in machine learning gaining attention because contemporary shielding materials urgently need optimization. This is because, conventional shielding approaches often face limitations in adaptability and material efficiency. Shi et al. (2022), and Elbaz et al., (2023) underscored the need for a thorough investigation into the integration of ML algorithms for tailoring shielding properties. Moreover, a comprehensive review is essential to pinpoint specific challenges and unexplored avenues in this evolving domain.

Through effective synthesis of information from diverse studies, this review aims to provide a clear understanding of fundamentals of radiation shielding, principles of radiation shielding and exposure, materials and traditional approaches to shield design, reinforced composites in shielding applications, ML in reinforced composite shield design, early adoption of ML in shield design, transition from rule-based to data-driven approaches, utilization of extensive datasets in shield optimization, algorithms for data-driven optimization, predictive modeling for shielding material properties, advanced material discovery, ML models for predicting shielding efficacy, training datasets and validation strategies, comparative analysis with traditional predictive models, scientific advancements in ML-assisted shield design, data preprocessing techniques, choice of ML algorithms, validation strategies and model interpretability, interdisciplinary collaboration in shield design, collaborative approaches in ML applications, ethical considerations in ML-assisted shield design, challenges and future directions, emphasizing the gaps that require further exploration. The synthesis of these findings

will contribute to refining ML-assisted design methodologies for reinforced composite radiation shields, fostering advancements in both materials' science and radiation protection.

This manuscript provided several important contributions to the field.

- The manuscript provides a detailed survey of radiation shielding principles, materials, and traditional approaches which provides the necessary groundwork for how machine learning can transform this field.
- It highlights the use of reinforced composites in radiation shielding applications with emphasis on their advantages over conventional materials, such as improved strength and radiation absorption.
- The manuscript searches how machine learning has been integrated into composite shield design with a special focus on predictive modeling and data-driven optimization to enhance shielding material properties and performance.
- It further tracks the early adoption of machine learning and the evolution from rule-based to more complex data-driven approaches, demonstrating the paradigm shift in the field.
- The review also discussed algorithms, predictive modeling, and training strategies, focusing on the role of data-driven techniques in optimizing shield material properties and efficacy.
- At the end, the manuscript concludes by outlining key challenges, such as data availability, model generalization, and the need for interpretable models, while advocating for ethical considerations in model development and data use.

The manuscript is organized into several well-structured sections, each addressing key aspects of the topic. It begins with Sect. 2 on Methodology, where the authors outline the research approach, including the techniques and frameworks used to investigate machine learning's role in shield design. Section 3 provides the Fundamentals of Radiation Shielding, offering a solid foundation for understanding the physics and materials involved in radiation protection. Following this, Sect. 4 discusses Materials and Traditional Approaches to Shield Design, presenting conventional methods and materials used in shielding. Sections 5 through 7 explore the integration of Reinforced Composites in shielding applications and the Early Adoption of Machine Learning in shield design. From Sect. 8 to 20, the manuscript covered the Transition to Data-Driven Approaches, discussing Machine Learning Algorithms, Predictive Modeling, and Training Datasets, offering a comparison with traditional methods. The manuscript concludes with a discussion of Challenges and Future Directions for machine learning in shield design, providing understanding into ongoing advancements and research opportunities in the field.

2 Methodology

The systematic literature review aimed to comprehensively examine research advancements in the field of ML-assisted design of reinforced composite radiation shields. The research questions guiding this review focused on recent breakthroughs, contributions of ML algorithms to shielding material optimization, and key challenges in integrating ML with radiation shield design. In establishing inclusion criteria, studies published from 2020 to the present were considered, encompassing research articles, conference papers, and reviews specifically addressing ML applications in reinforced composite radiation shield design. Exclusion criteria included research not written in English or those irrelevant to the ML aided design of radiation shielding materials, as well as literature distributed preceding 2020.

The search procedure included an orderly investigation of major scientific databases, including PubMed, IEEE Xplore, ScienceDirect, and Google Scholar. Keywords and search terms, for example, " "ML," "reinforced composite," and "radiation shielding," were decisively consolidated to refine search inquiries, using Boolean administrators for ideal outcomes. For study selection, two free commentators embraced a two-stage process. At first, titles and digests were evaluated for significance to the exploration questions. In the resulting stage, full texts of chosen studies were investigated to decide last consideration. Any inconsistencies were settled through conversation and, when strategic, interview with a third commentator.

Data extraction included the utilization of a normalized structure to gather relevant data from the included examinations systematically. This incorporated subtleties like review attributes (e.g., author, publication year), ML algorithms employed, reinforcement materials, radiation types considered, key discoveries, and constraints. A narrative synthesis approach was picked for data union, considering the outline and integration of discoveries from the included investigations. This approach intended to distinguish normal subjects, patterns, and achievements connected with ML-assisted design of reinforced composite radiation shields.

3 Fundamentals of radiation shielding

Radiation encompasses various forms of energy transmission, categorized into ionizing and non-ionizing types. Ionizing radiation includes alpha and beta particles, emitted during radioactive decay, and gamma rays, a high-energy electromagnetic radiation (Karmaker et al.,

2022). Non-ionizing radiation includes microwaves, radiofrequency waves, infrared radiation, visible light, ultraviolet radiation, extremely low-frequency radiation, and X-rays (Maqbool, 2023). Neutron radiation consists of neutral particles released from nuclear reactions (Leyva and Munévar 2023). While ionizing radiation can potentially damage biological tissues due to its ability to ionize atoms, non-ionizing radiation is generally considered less harmful. Neutron radiation and X-rays have specific applications in various fields, from medical imaging to industrial processes.

Ionizing radiation, an unavoidable power in our current circumstance exuding from regular and human-made sources, has been a point of convergence of scientific request, starting an overly complex investigation into its likely hazards and mitigating methodologies. At its center, ionizing radiation gets from the capacity to free electrons from molecules, a peculiarity that can bestow energy to encompassing matter (Sobczak and Żyła, 2024; Bassam et al. 2024; Hamisu et al. 2023). The natural world, with its cosmic rays and terrestrial radiation arising from elements in soil and rocks, contributes significantly to background radiation. The literature converges on the consensus that while these natural sources are inherent in our daily lives, it is the anthropogenic origins of ionizing radiation that warrant meticulous scrutiny. Industries such as nuclear power generation (Iqbal et al. 2021), medical applications (Omer 2021), and research endeavors (Chauhan et al. 2022) are cited as notable contributors to the anthropogenic footprint, necessitating a detailed examination of their impact on human health and the environment.

A fundamental distinction emerges within the literature, demarcating the hazards associated with natural and man-made sources of ionizing radiation. Natural sources, though omnipresent, are generally deemed low risk (Liu et al. 2022a; Baroň et al. 2022). It is the exercises connected to human intercession, for example, thermal energy stations, clinical diagnostics (Lu et al. 2021), and modern applications, that drive the talk into domains of concern. The literature highlights the basics of figuring out some kind of harmony between receiving the rewards of these innovative progressions and alleviating the potential hazards they present.

The dangers related to ionizing radiation are not uniform, dependent upon factors like dose, duration of exposure, and the specific type of radiation involved. High doses can prompt intense impacts, appeared as radiation affliction and organ harm, while delayed exposure to bring down dosages hoists the risk of disease and hereditary changes (Wang et al. 2023). The literature explores through the complex landscape of dose reaction interactions, disclosing the fragile balance between the benefits and dangers inborn in ionizing radiation.

Inside the domain of clinical applications, ionizing radiation expects an urgent role in diagnostic imaging and cancer therapy. X-rays (Zhu et al. 2023), Computed Tomography (CT) scans, and radiotherapy (Siddique et al. 2023) stand as innovative achievements, changing clinical diagnostics and treatment regimens. The literature bears witness to the fundamental nature of these modalities while simultaneously recognizing the basic to take on systems for dose optimization and mechanical advancements to curtail unnecessary exposure. Scientists and material designers have participated in a ceaseless quest for innovation, maximizing therapeutic efficacy and minimizing the potential for harm. The literature typifies an aggregate work to overcome any issues between scientific data and public comprehension. Risk correspondence arises as a repetitive subject, recognizing the basic to cultivate informed navigation and dissipate misinterpretations encompassing ionizing radiation.

3.1 Principles of radiation shielding and exposure

Radiation shielding remains as a foundation in the domain of nuclear safety and radiological protection, a subject complicatedly woven into the texture of scientific literature. The strategic principles of radiation shielding are established in the physical science of ionizing radiation and the basic to limit the exposure of people and the environment to its possibly destructive impacts. Radiation shielding principles, known as ALARA (Baker and Kamboj 2022; Ambrose 2023), emphasize minimizing exposure by managing time, distance, and shielding as shown in Fig. 2. The arrow in Fig. 1 signifies progression and sequence, illustrating that effective radiation safety involves a combination of strategies ranging from reducing exposure time, increasing distance from the source, to applying effective shielding. It emphasizes the interconnected and cumulative nature of these measures to minimize radiation exposure comprehensively. Reducing the duration of exposure (time) is important, achieved through efficient work practices. Increasing the distance from radiation sources is key, as intensity diminishes with the square of the distance (Azman et al. 2023).

Shielding materials, like lead or concrete, serve as barriers, absorbing or attenuating radiation. The goal is to lower exposure to levels that are reasonably achievable and safe. The literature combines on the comprehension that radiation shielding works on the important reason of attenuation, the decrease in force as radiation crosses matter. This attenuation is dependent upon elements like the sort and energy of radiation, the density and density of the shielding material, and the distance between the radiation source and the focal point (Luoni et al. 2022; Sayyed 2024).

A basic perspective featured in the literature is the differentiation between direct and indirect shielding. Direct shielding includes putting a defensive boundary between



Fig. 1 Principles of Radiation shielding and exposure

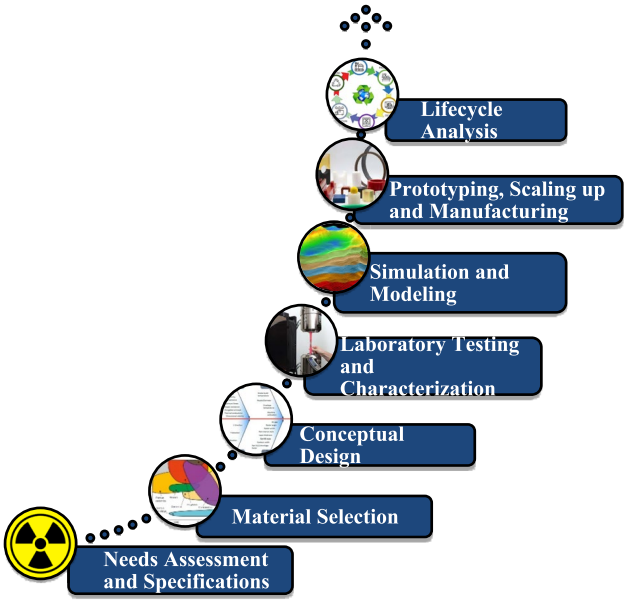


Fig. 2 Stages of reinforced material design for radiation shielding purpose

the radiation source and the target, while indirect shielding entails creating a barrier that absorbs and scatters radiation within the immediate vicinity of the source (Lakshminarayana et al. 2020). The literature highlights the significance of choosing shielding materials that line up with the qualities of the radiation source, utilizing a detailed way to maximize efficacy.

The different nature of ionizing radiation requires a custom fitted methodology, considering the energy range and sort of radiation experienced (Kim 2021). Traditional shielding materials incorporate lead (Elsafi et al. 2022; El-Khatib et al. 2023), cement (Amin et al. 2023), and steel (Oh et al. 2023), each having novel attributes that render them reasonable for explicit applications. The literature explores through the unpredictable equilibrium of material properties, like density and atomic number, in deciding the shielding viability against various kinds of radiation. However, advancements in materials science deliver novel composites and amalgams, extending the collection of choices accessible for enhancing shielding techniques.

The literature reviewed reveals insight into the significance of complete design considerations in radiation shielding. The spatial game design of shielding materials, alongside factors like density (Sayyed 2023) and design (Zheng et al. 2023), assumes a vital part in deciding the general viability of the shielding barrier. Design principles reach out past the actual design, including considerations of inhabitance, utilization examples, and potential situations that might affect the adequacy of the shielding measures. The literature underscores the combination of safety edges and overt repetitiveness in design to represent unexpected conditions and guarantee a vigorous shielding against radiation dangers (Tamam et al. 2024).

The dynamic nature of radiation shielding is illuminated through the exploration of emerging technologies and innovative approaches in literature. Computational tools and simulations, highlighted in the literature, facilitate the design and optimization of shielding strategies, offering a virtual sandbox to assess the efficacy of different materials and designs. Besides, the literature uncovers progressing investigation into versatile shielding frameworks that can powerfully conform to changing radiation levels, giving a responsive and fitted way to deal with the difficulties presented by fluctuating radiation conditions (Laanen et al. 2023).

4 Materials and traditional approaches to shield design

Traditional ways to deal with shield design have for quite some time been moored in the basic principles of physical science, zeroing in on the weakening of ionizing radiation through the strategic utilization of materials. The literature synthesizes a wealth of studies that collectively illuminate the conventional wisdom embedded in shield design methodologies. The overall objective has reliably been to limit the exposure of people and the environment to possibly

destructive radiation (Okonkwo et al. 2022), and traditional methodologies have laid the basis for understanding and addressing this goal. The exploration of materials employed in radiation shields constitutes a critical facet within the corpus of scientific literature, offering a systematic examination of the diverse substances harnessed to attenuate and mitigate the effects of ionizing radiation.

Lead, a venerable stalwart in radiation shielding, emerges as a focal point in the literature (Elsafi et al. 2022). The inherent high density and atomic number of lead render it exceptionally effective in attenuating gamma rays and X-rays. Studies within the literature systematically quantify the shielding capacity of lead, delving into parameters such as density and design to optimize its effectiveness (Khosravi et al. 2023). While lead excels in certain applications, the literature critically appraises its drawbacks, including toxicity and environmental concerns (Boukhris et al. 2021). Researchers and practitioners alike have historically leaned on lead as a reliable and readily available material, forging a conventional path in shield design (Rudychev et al. 2021). Researchers advocate for a judicious approach to lead usage, balancing its shielding prowess with the imperative to explore alternative materials that mitigate environmental impact.

Concrete surfaces as another stalwart in the literature, particularly in the context of nuclear power designs and medical facilities. The composite nature of concrete, with its mixture of aggregate and cement, confers substantial shielding capabilities against both gamma and neutron radiation (Ban et al., 2021). The systematic literature review encapsulates studies that meticulously evaluate the density and composition of concrete, unraveling the intricate relationship between these variables and the resultant shielding effectiveness. While concrete offers advantages in terms of cost and ease of fabrication, the literature underscores the need for ongoing research to enhance its performance and address limitations, such as its comparatively lower efficacy against high-energy gamma rays (Wang et al. 2023).

Steel, renowned for its durability and malleability, surfaces as a versatile contender in radiation shielding. The literature systematically dissects the properties of steel, exploring its utility in diverse applications ranging from medical facilities to industrial settings (Kassab et al., 2023). The magnetic properties of steel also come under scrutiny, with studies investigating their potential impact on shielding efficacy. The systematic exploration within the literature converges on the versatility of steel, while simultaneously emphasizing the importance of selecting the appropriate alloy and density to optimize shielding performance (Zhang et al. 2022a, b).

Beyond traditional materials, the literature systematically unravels the promise held by novel composites and alloys in advancing radiation shielding technology. Borated polyethylene, for instance, emerges as a composite material with unique neutron-absorbing properties, garnering attention in nuclear reactor shielding applications (Khosravi et al. 2023). The systematic literature review encapsulates studies that systematically investigate the composition, density, and structural characteristics of these innovative materials, providing a comprehensive assessment of their viability in diverse radiation shielding scenarios (Shahzad et al. 2022). The literature systematically examines the alignment between regulatory mandates and the practical implementation of shielding materials. It underscores the imperative for a systematic approach to compliance, emphasizing the need for materials that not only meet regulatory benchmarks but also align with sustainability and environmental responsibility.

5 Reinforced composites in shielding applications

The interaction of radiation with different materials is a key idea that assumes a strategic part in grasping the properties and behavior of various substances. This interaction envelops a wide range of peculiarities and has significant ramifications in fields like materials science, nuclear engineering, and natural science. The principles of radiation interaction with matter incorporate many peculiarities, and they have significant ramifications for different applications, including nuclear physical science, clinical imaging, and radiation treatment (Oksengendler et al. 2023). Radiation incorporates an expansive range of peculiarities, including electromagnetic radiation (e.g., X-rays and gamma rays) and molecule radiation (e.g., alpha particles, beta particles, and neutrons). These kinds of radiation have various properties, like charge, mass, and energy, which impact their interactions with matter (Pathak2023). One significant differentiation in radiation collaboration is among ionizing and non-ionizing radiation. Ionizing radiation has adequate energy to eliminate firmly bound electrons from particles or atoms, prompting the formation of particles (Bassam et al. 2024). Non-ionizing radiation, then again, comes up short on energy expected for this cycle (Nyakyi et al. 2024; Galdino et al. 2023). X-rays, gamma rays, and certain particles like alpha and beta particles are ionizing, while noticeable light, radio waves, and microwaves are non-ionizing.

Material designers are concerned about the attenuation of ionizing radiation. The degree of weakening is impacted by the energy of the radiation, the density of the material, and its atomic composition (Mostafa et al. 2020; Sobczak & Żyła, 2024). For example, higher-density materials and higher-energy radiation bring about more noteworthy attenuation. The likelihood of absorption relies upon the sort of radiation and the nature of the material. The organization of

a material strategically impacts how it cooperates with radiation. Different elements and compounds have varying atomic and molecular structures, which affect their responses to incident radiation (Matsuya & Saga 2023). For example, materials with high atomic numbers tend to be more effective at attenuating ionizing radiation due to increased electron density.

Whether it's shielding workers in nuclear offices or travelers in the air, materials should weaken and dissipate incoming radiation while keeping up with primary respectability. Reinforced composite materials have become fundamental in many enterprises because of their excellent mechanical properties, lightweight nature, and protection from corrosion. These materials comprise of a matrix material and reinforcing fibers, and they are utilized in applications going from aviation and car to development and clinical gadgets (Vinayagamoorthy 2022). One basic part of composite material design is the capacity to tailor their electromagnetic shielding properties. The shielding adequacy of reinforced composite materials is accomplished through a few mechanisms like reflection, absorption, scattering, different reflections, and absorption by composing elements. Hence, inclusion of high-quality reinforcements that supports this mechanism in "right" proportion is paramount in achieving effective attenuation.

In composites, the choice of materials with high reflectivity can increase the shielding effectiveness. Electromagnetic waves that penetrate the shielding material are absorbed and converted into heat energy. This mechanism is important in shielding materials designed for high-frequency electromagnetic radiation (Jia et al. 2022; Park et al. 2021). Through effective incorporation of scattering centers into composite materials, electromagnetic waves can be redirected or attenuated. The ability of a reinforcement or combination of reinforcements to achieve multiple reflections is strategic since each reflection results in a reduction of radiation intensity. Multiple reflections occur when electromagnetic waves pass through a composite and reflect between interfaces within the material. In addition, the inclusion of conductive fibers or fillers in the composite can enhance shielding (Morales et al. 2023; Wang et al. 2021). Conductive elements create paths for the absorption and dissipation of electromagnetic energy, reducing the radiation that can penetrate the material.

The task of achieving "right" proportion and combination of reinforcements within a plastic or polymeric matrix presents several challenges. This is because choosing the appropriate materials for the matrix and reinforcing elements is a complex process (Qiao et al. 2023). It involves a trade-off between the material's inherent electromagnetic properties, mechanical strength, and weight considerations. Accomplishing the ideal shielding adequacy at explicit electromagnetic frequencies and for various sorts of radiation (e.g., radiofrequency, microwave, or X-ray) is challenging. Traditional trial-and-error approaches frequently lead to exorbitant and tedious iterations. The game design of reinforcing strands inside the matrix, known as the composite's microstructure, altogether affects its shielding performance. Moreso, determining the optimal microstructure for a given application is challenging without efficient design methods (Tan et al. 2020). Therefore, accurate prediction of shielding effectiveness necessitates complex numerical simulations that consider the interactions between electromagnetic waves and composite structures. These simulations can be computationally intensive and time-consuming without ML tools.

6 Machine learning in reinforced composite shield design

The evolution of ML in shield design represents a paradigm shift, challenging and expanding upon traditional approaches. The literature synthesizes a diverse array of studies (Li et al. 2022; Shi et al. 2022; Sreenivas et al. 2023; Boahen et al. 2023a), revealing the progressive adoption of ML algorithms to optimize the design, and material selection for radiation shields. The systematic literature analysis unfolds, capturing the essence of how ML has become an integral force reshaping the landscape of radiation protection (Dorise et al. 2021). Studies underscore the capacity of ML to discern intricate patterns and relationships within vast datasets, heralding a new era in shield design. The adaptive nature of ML algorithms enables them to autonomously learn and refine strategies for shielding against ionizing radiation, transcending the limitations of traditional, rule-based methodologies. The systematic lens applied to the literature unveils the potential of ML to revolutionize the precision, efficiency, and adaptability of shield design (Dercle et al. 2021; Sreenivas et al. 2023).

Materials, long considered the building blocks of shields, undergo a transformation in the wake of ML integration. The literature systematically dissects studies where ML algorithms optimize the selection of materials based on their unique properties and interactions with different radiation types (Shi et al. 2022). This evolution in material selection represents a departure from traditional reliance on trial and error, ushering in an era where ML-driven strategies embrace a diverse array of materials tailored to specific radiation scenarios. The systematic exploration extends to spatial arrangement and design, as ML algorithms contribute to the design optimization process (Liu et al. 2022b; Shi et al. 2022). The studies under review depict how ML models analyze complex spatial relationships, suggesting innovative designs that enhance shielding effectiveness (Shi et al. 2023a, b). The systematic approach allows for a granular examination of how ML augments traditional

design principles, fostering a dynamic interplay between data-driven understanding and established practices.

ML's evolution in shield design extends beyond conventional paradigms, delving into predictive modeling and optimization (Bilmez et al. 2022; Boahen et al. 2023b; Sreenivas et al. 2023). The literature systematically unfolds studies where ML models forecast radiation exposure levels, offering valuable understanding for preemptive design strategies. Optimization algorithms driven by ML aim to identify designs and materials that not only meet regulatory principles but also push the boundaries of efficiency and efficacy.

Through effective training of models on vast datasets of simulated and experimental results, ML algorithms can provide rapid and accurate predictions of shielding performance. This enables designers to iterate through multiple design options quickly (Morgan et al. 2022; Boahen et al. 2023a). ML can extract valuable information from experimental data and simulations that may not be immediately apparent through traditional analysis (Boahen et al. 2023b; Malidarre et al. 2022). This data can guide the design process by revealing the most critical factors influencing shielding effectiveness. ML techniques have the capacity to enhance the design and assessment of radiation shielding materials in terms of material selection, tailoring shielding performance, predicting shielding effectiveness, and data-driven analysis.

6.1 Machine learning deployment in various stages of reinforced composite design for radiation shielding purpose

Designing reinforced materials for radiation shielding purposes is a complex and critical process, typically involving several stages to ensure the materials effectively attenuate ionizing radiation. These materials are strategic for shielding individuals, equipment, and the environment from the harmful effects of radiation in various applications. The stages involved in the design of reinforced materials for radiation shielding are summarized in Fig. 2:

6.1.1 Needs assessment and specifications stage

The needs assessment and specifications stage serve as the foundational cornerstone in the design of reinforced composite shielding materials, laying the groundwork for subsequent processes. ML plays a transformative role in enhancing the efficacy and precision of this phase (Bilmez et al. 2022; Sreenivas et al. 2023). It engages in comprehensive data analysis, extracting understanding from historical data repositories, research publications, and industry reports. In fields like medical radiography, ML mines data to understand the energy spectrum of commonly used X-rays (Yang et al. 2021a, b).

ML recognizes patterns within historical data, guiding the discovery of recurring patterns associated with radiation exposure. This predictive capability allows for anticipating the types and energy levels of radiation that the material should shield against. Risk assessment is a profound application, where ML evaluates potential risks related to radiation exposure by considering attributes such as radiation type, energy levels, exposure durations, and the presence of humans or sensitive equipment. These assessments contribute significantly to the precise delineation of material specifications. Another important role of ML is in property prediction (Kaşkaş and Şahmaran 2022). Tailored predictive models predict the radiation attenuation properties required for new composite materials based on historical data characterizing materials with known properties and their corresponding shielding effectiveness. In essence, ML acts as a beacon, optimizing the needs assessment and property specification process, guiding informed choices in the design of shielding materials.

6.1.2 Material selection stage

The Material Selection stage in developing reinforced composite materials for radiation shielding is important, demanding precision in choosing materials, compositions, and designs. ML transforms this phase, introducing a data-driven approach that enhances choice-making efficiency (Shi et al. 2022). Traditionally, material selection relied on experience and norms, but ML leverages historical data from various sources, compiling extensive datasets on material properties under ionizing radiation. Linear attenuation coefficients, mass attenuation coefficients, and half-value layers (HVLs) are key attributes assessed to determine material suitability (Hassanpour et al. 2023).

ML models, trained on historical data, predict radiation attenuation properties and recommend optimal material compositions, considering factors like density, cost, and mechanical strength (Boahen et al. 2023a, b). These algorithms act as expert guides, providing application-specific recommendations for diverse scenarios. In medical radiography, they may suggest materials minimizing X-ray exposure, while for the nuclear industry, they recommend comprehensive protection from gamma radiation. ML's adaptability and iterative nature enable swift updates with new data, ensuring agile material selection that addresses evolving radiation sources, energy levels, and usage scenarios. This transformative role positions ML as a cornerstone in optimizing the Material Selection stage for reinforced composite shielding materials.

6.1.3 Conceptual design stage

The conceptual design stage in developing reinforced composite shielding materials is a pivotal moment where creative solutions are conceived to meet the demands of ionizing radiation protection. ML's integration in this phase marks a transformative era, leveraging data-driven understanding and predictive modeling to optimize solutions. This phase is the birthplace of innovative ideas, framing the trajectory for subsequent stages (Ampanavos et al. 2021). ML's role in data analysis allows researchers to mine historical data, research papers, and industry reports, providing empirical knowledge of material performance under ionizing radiation. Pattern recognition within historical data serves as inspiration for creative ideation, guiding researchers to envision novel materials or structures for superior radiation attenuation.

ML extends its influence on feasibility assessment, gauging each creative idea's viability based on factors like material availability, manufacturing processes, and cost considerations. ML filters out impractical or expensive concepts, narrowing down potential ideas for realization. ML's adaptability ensures that these algorithms evolve with changing data, providing updated assessments of feasibility and potential performance. This responsiveness to the latest data allows for the incorporation of current understanding into the material design process, enhancing the conceptual design stage's effectiveness.

6.1.4 Laboratory testing and characterization stage

The laboratory testing and characterization stage is a critical stage in the improvement of reinforced composite shielding materials, where materials are orchestrated, shaped, and exposed to thorough testing to meet explicit shielding performance necessities. ML integration transforms this stage, improving accuracy and productivity through data-driven bits of knowledge and predictive modeling (Aggarwal et al. 2021). In the laboratory, materials undergo detailed examination, assessing properties like radiation attenuation coefficients, mechanical strength, thermal stability, and density (Al-Hassan et al. 2021). Test findings shape material composition and design to meet radiation attenuation and safety requirements.

ML's impact begins with material synthesis, providing recommendations based on historical data and predictive models for reinforcing elements, proportions, and manufacturing processes. This data-informed foundation ensures a more effective synthesis process (Wu et al. 2024). In material characterization, ML predicts measured properties, such as radiation attenuation and mechanical strength, based on historical data, streamlining testing, and offering valuable context. Additionally, ML optimizes the testing process by recommending suitable methods and procedures, focusing on critical properties for radiation shielding. The mix of ML in the Laboratory Testing and Characterization stage denotes a huge advancement, empowering a more exact, productive, and data-driven approach in the improvement of reinforced composite shielding materials (Guo et al. 2021).

6.1.5 Simulation and modeling stage

The simulation and modeling stage assumes a significant part in the improvement of reinforced composite shielding materials, going about as the interaction between hypothetical design and practical implementation. In this stage, scientists utilize different ML models to predict material behavior under unambiguous radiation conditions, expecting to advance arrangements and designs (Bhamare et al. 2021; Ford et al. 2021). The integration of ML transforms this stage by leveraging data-driven understanding to enhance precision and efficiency in material development. This phase represents virtual trial-and-error, employing complex mathematical models and simulations to predict shielding material performance.

ML in the simulation and modeling stage starts with assimilating and analyzing historical data, including material performance under various radiation types and exposure conditions (Dastider et al. 2021). ML models mine this data for patterns and understanding, providing valuable context for engineering more effective shielding materials. Notably, ML predicts material behavior in radiation environments by training models on historical data to forecast attenuation, considering factors like linear attenuation coefficients and mass attenuation coefficients (Varone et al. 2023; Amin et al. 2022). These predictions inform the refinement of material composition and structure, optimizing radiation shielding properties. The incorporation of ML in this stage signifies a transformative shift, introducing a data-driven approach to enhance the precision and effectiveness of simulations in the development of reinforced composite shielding materials.

6.1.6 Prototype development stage

The prototype development stage in creating reinforced composite shielding materials is an important phase where theoretical designs and simulations are translated into real-world prototypes. The integration of ML in this stage enhances models, epitomizing designs from prior stages, act as physical testbeds to approve hypothetical ideas, portray material properties, and guarantee consistence with stringent radiation attenuation necessities (Okafor et al. 2023). ML in the prototype development stage begins with assimilating and analyzing historical data, providing valuable context for prototype development. This data, encompassing material properties under diverse radiation conditions, serves as a foundation for constructing effective shielding materials. ML's

predictive capabilities are critical, forecasting how prototypes will perform under various radiation conditions (Morgan et al. 2022). The adaptability of ML models is evident, adjusting predictions and recommendations as new data and understanding emerge, ensuring prototypes align with the latest data. The infusion of ML in the Prototype Development stage signifies a transformative shift, enabling a more responsive and precise approach to material development.

6.1.7 Scaling up and manufacturing stage

The scaling up and manufacturing stage in reinforced composite shielding materials implies the transition from laboratory prototypes to large scale manufacturing, requiring steady, effective assembling while at the same time satisfying severe quality guidelines. The combination of ML in this stage presents a data-driven approach, utilizing verifiable data, predictive modeling, and versatility to upgrade proficiency, quality control, and optimization (Hasti et al. 2021). As researchers move from limited scope models to large scale manufacturing, they face the test of guaranteeing versatility, unwavering quality, and cost-adequacy without compromising radiation attenuation properties. ML becomes instrumental by smoothing out assembling cycles and quality control through data-driven experiences (Han and Kwon 2021). Historical data, encompassing material properties, manufacturing processes, and quality control measures, serves as a foundation for ML models, extracting patterns and understanding for a data-informed production approach.

ML's notable application in this stage includes process optimization, recommending efficient and cost-effective manufacturing processes based on historical data and predictive modeling. These recommendations consider the material's composition, structure, and quality requirements, ensuring streamlined and optimized production. Quality control is another important aspect where ML predicts expected quality attributes based on historical data and the manufacturing process (Merchant et al. 2023). The incorporation of ML in the scaling up and manufacturing stage transforms the production process, enhancing efficiency, quality, and optimization in the mass production of reinforced composite shielding materials.

6.1.8 Lifecycle analysis stage

The lifecycle analysis stage is the final phase in reinforced composite shielding material development, focusing on assessing sustainability, performance, and environmental impact throughout the materials' lifecycle. ML introduces a data-driven approach, leveraging historical data, predictive modeling, and adaptability to enhance sustainability, performance optimization, and choice-making (Ji et al. 2021). This stage involves a comprehensive examination of materials throughout their lifecycle, considering performance, environmental impact, energy efficiency, and production and maintenance costs. The challenge is to ensure materials remain effective and sustainable while minimizing environmental impact. ML is instrumental in this assessment, enabling a holistic and data-driven approach.

ML's role in the lifecycle analysis stage begins with assimilating and analyzing historical data related to material performance, maintenance, and environmental impact. The models mine this data for patterns and understanding, forming the foundation for lifecycle analysis (Yang et al. 2021a, b). Historical data allows researchers to evaluate long-term performance and sustainability based on empirical evidence. Predictive modeling for performance optimization is a key application of ML in this stage. Models forecast material performance over time under various radiation conditions, informing choices about maintenance, replacement, or refurbishment to optimize longevity. Environmental impact assessment is another critical aspect, predicting consequences of production, usage, and disposal based on historical data and predictive modeling. This informs choices on sustainable practices and reducing environmental impact, contributing to long-term material sustainability.

7 Early adoption of machine learning in shield design

The early adoption of machine learning in shield design has proclaimed a groundbreaking period in the field of radiation shielding. As machine learning keeps on advancing, its application in shield design is ready to reclassify the way we make viable safety against ionizing radiation (Sreenivas et al. 2023). Historically, shield design was a labor-intensive, empirical process that heavily relied on manual calculations and testing. Researchers and engineers would manually assess the properties of various materials and designs, often based on theoretical knowledge and experience. While viable somewhat, this traditional methodology had limitations concerning accuracy and flexibility to changing radiation situations.

The early reception of ML achieved a change in outlook by imbuing data-driven experiences into the design iteration. Through effective investigation of authentic information on material properties, radiation types, exposure levels, and viability, ML models can forecast the shielding adequacy of various materials and designs (Scott et al., 2021). This predictive ability engages designers to settle on informed conclusions about the choice and advancement of shielding materials. For instance, in clinical radiography, ML can predict how different materials will weaken X-rays, considering the customization of shielding arrangements in view of

the radiographic hardware and exposure conditions (Varone et al. 2023; Bilmez et al. 2022).

Moreover, ML presents a degree of flexibility that was beforehand out of reach. Radiation sources and exposure situations are dynamic and can change after some time. ML models can change their forecasts and proposals as new information opens, guaranteeing that shield designs stay powerful and exceptional (Bhosekar & Ierapetritou 2020). This adaptability is particularly vital in fields like nuclear energy and space exploration, where radiation conditions may vary widely. One of the hallmark applications of ML in shield design is in the optimization of material composition (Chen et al. 2020). ML also contributes to the design of novel shielding materials. Through effective analysis of material properties and performance under radiation exposure, ML models can suggest the development of new composite materials with enhanced radiation attenuation properties. This innovation-driven approach has the potential to yield groundbreaking materials that outperform traditional options.

8 Transition from rule-based to data-driven approaches

Traditionally, shield design adhered to rule-based methodologies, governed by established principles and empirical rules. In this paradigm, key features of rule-based approaches, as synthesized in Table 1, are evident. Material selection hinged on predetermined rules such as the density and atomic number, leading to common choices like lead for gamma rays. Designs were strictly predetermined, guided by empirical guidelines rather than adaptive strategies (Li et al. 2023). The spatial arrangement, important for effective shielding, followed rule-based principles derived from established shielding concepts (Sun et al. 2022). This structured approach, while providing a systematic foundation, lacked the adaptability and precision inherent in more contemporary, data-driven methodologies. The transition from these rule-based strategies to more dynamic approaches reflect an evolution toward optimizing shielding efficacy through a deeper understanding of materials and dynamic spatial considerations.

The literature review reveals a transformative paradigm shift towards data-driven approaches in shield design, driven by remarkable advancements in ML and computational capabilities. Table 1 presents a comprehensive

Table 1 Characteristics of data-driven and key features of rule-based approaches in shield design

Aspect	Data-Driven Approach	Rule-Based Approach	Source
Material Selection	ML algorithms optimize material selection based on complex interactions with radiation types	Based on established density and atomic number rules (e.g., lead for gamma rays)	Zhu et al. 2021
Design	Adaptive algorithms suggest optimal designs through analysis of spatial relationships	Predetermined designs derived from empirical guidelines	Sharma et al. 2022a, b
Spatial Arrangement	Data-driven understanding guide spatial arrangements, considering dynamic and adaptive strategies	Rule-based spatial arrangements influenced by shielding principles	Sun et al. 2022
Cost Optimization	ML models analyze cost-effectiveness for material and design choices	Rule-based approaches may consider cost, but decisions are often based on predefined guidelines	Schoenholz and Zobeiry 2024; Sharma et al. 2022a, b; Shi et al. 2022
Adaptive Updates	Data-driven approaches can adapt to new data and update shielding strategies dynamically	Rule-based approaches may require manual updates based on changing guidelines or standards	Chen et al. 2023
Simulation Accuracy	Utilizes simulations and real-world data for accurate predictions and optimizations	Relies on predefined rules, which may not capture all nuances of complex radiation interactions	Sidi-Salah et al. 2020
Customization	Can provide highly customized solutions based on specific facility requirements	Rule-based approaches may lack the flexibility to cater to unique or complex scenarios without extensive rule modifications	Song et al., 2022; Yang et al. 2023
Sustainability	Optimization considering environmental impact, incorporating green materials and energy-efficient designs	Rule-based approaches may not inherently prioritize sustainability without explicit considerations in predefined rules	Tsaples et al. 2022

overview of characteristics in data-driven and rule-based approaches to shield design, highlighting key aspects such as material selection (Zhu et al. 2021), design (Sharma et al. 2022a, b), spatial arrangement (Sun et al. 2022), cost optimization (Schoenholz and Zobeiry 2024), adaptive updates (Chen et al. 2023), simulation accuracy (Sidi-Salah et al. 2020), customization (Yang et al. 2023), and sustainability (Tsaples et al. 2022). In the data-driven approach, ML algorithms optimize material selection based on intricate radiation interactions, suggest adaptive designs through spatial relationship analysis, and provide dynamic spatial arrangements. Additionally, this approach excels in cost optimization, adaptive updates, and customization, offering tailored solutions for specific facility needs. It prioritizes sustainability by optimizing for environmental impact and incorporating green materials. In contrast, the rule-based approach relies on predetermined rules, demonstrating proficiency in established density and atomic number guidelines for material selection, and providing predetermined designs based on empirical guidelines. However, it may lack the adaptability and customization of data-driven approaches and may not inherently prioritize sustainability without explicit considerations.

The literature underscores the implications and advantages associated with the transition to data-driven approaches. The transition from rule-based to data-driven approaches in shield design carries profound implications that redefine the landscape of radiation protection. The implications of this transition include enhanced efficacy in shielding strategies, as ML excels in recognizing intricate patterns within radiation data. Real-time adaptability becomes a cornerstone, enabling shields to dynamically respond to evolving radiation scenarios. The optimization of shielding strategies, a key outcome of data-driven approaches, ensures not only enhanced protection but also increased efficiency in the utilization of shielding materials. As these implications unfold, the transition heralds a paradigm shift, offering a detailed, dynamic, and data-driven approach to shield design that aligns with the evolving complexities of radiological environments.

9 Utilization of extensive datasets in shield optimization

The evolution of shield optimization is intricately tied to the utilization of extensive datasets, ushering in a paradigm shift in the approach to radiation protection. Table 2 illuminates the key aspects of utilizing extensive datasets in shield optimization, offering a comprehensive overview of the transformative role of data-driven approaches. Material properties benefit from extensive datasets, becoming a rich source for informed choices in shielding material selection. Radiation characteristics, encompassing diverse sources and energy levels, are optimized with detailed precision through dataset-driven methodologies. Refining computational models, a important aspect, is achieved with the aid of extensive datasets, ensuring simulations that are accurate and reliable. The inclusion of environmental considerations speaks to the commitment to sustainability, as datasets inform assessments of the environmental impact of shielding materials. Additionally, cost–benefit analyses leverage extensive datasets, guiding choices to achieve optimal economic efficiency in the intricate realm of shield design. Collectively, these aspects underscore the multidimensional advantages of incorporating extensive datasets, shaping the future of shield optimization with adaptability, precision, and sustainability at its core.

The literature systematically scrutinizes studies that leverage extensive datasets to inform material selection for shield optimization. Datasets encompassing diverse materials and their properties enable understanding of shielding efficacy. Table 3 delves into material-related considerations in shield optimization, showcasing the pivotal role of extensive datasets. Density and atomic number rules, informed by datasets, optimize material choices, enhancing shielding effectiveness against specific radiation types (Li et al. 2023). The assessment of toxicity and environmental impact, facilitated by datasets, underscores a commitment to sustainability in shield optimization strategies. Thermal and mechanical properties are meticulously addressed through comprehensive datasets, ensuring materials align with operational

Table 2 Key Aspects of Utilizing Extensive Datasets in Shield Optimization

Aspect	Utilization of Extensive Datasets in Shield Optimization	Source
Radiation Characteristics	Datasets encompassing a wide range of radiation characteristics enable detailed optimization of shields, considering diverse radiation sources and energy levels	Wu et al. 2024
Computational Modeling Parameters	Utilizing extensive datasets aids in refining computational models, ensuring accuracy and reliability in simulations for optimizing shield designs	Amin et al. 2022
Environmental Considerations	Datasets inform the assessment of the environmental impact of shielding materials, ensuring sustainability in shield optimization strategies	Cordella et al. 2024
Cost–Benefit Analysis	Extensive datasets contribute to cost–benefit analyses, guiding choices on material and design choices for optimal economic efficiency in shield design	Ciapessoni et al. 2023

Table 3 Material-Related Considerations and Dataset Utilization

Material Aspect	Dataset Utilization in Shield Optimization	Source
Density and Atomic number	Extensive datasets inform choices based on density and atomic number rules, optimizing shielding effectiveness against specific radiation types	Fetzer et al. 2024
Thermal and Mechanical Properties	Comprehensive datasets inform choices regarding the thermal and mechanical properties of shielding materials, ensuring compatibility with operational requirements	El-Mesady et al. 2023
Availability and Cost	Datasets contribute to considerations of material availability and cost, enabling an informed balance between efficacy and economic feasibility in shield design	Yuhang et al. 2024
Radiological Stability	Extensive datasets inform choices regarding the radiological stability of shielding materials, ensuring long-term effectiveness in radiation protection strategies	Luoni et al. 2022

requirements (Kocyigit et al., 2023). Material availability and cost considerations benefit from datasets, guiding an informed balance between efficacy and economic feasibility in shield design. Moreover, datasets contribute to choices regarding the radiological stability of shielding materials, ensuring their enduring effectiveness in the realm of radiation protection. In amalgamation, these material-related considerations emphasize the indispensable nature of datasets in crafting effective, sustainable, and operationally compatible shield optimization strategies.

The systematic analysis extends to the utilization of extensive datasets to optimize shields considering a spectrum of radiation characteristics. From gamma rays to neutron flux, diverse datasets allow for a comprehensive approach to shield design (Yuhang et al. 2024). Table 4 intricately explores radiation characteristics and their dataset-driven optimization in shield design. Utilizing datasets with data on energy levels and spectra serves as a guide, facilitating the optimization of shields tailored for specific radiation sources. The capture of angular distribution in datasets aids in the formulation of shields that effectively mitigate radiation from various incident angles, adding a dynamic dimension to their efficacy. Comprehensive datasets on interaction cross-sections play a important role in informing optimization strategies for shielding against specific radiation types, ensuring precision in design (Melia and Parsons 2023; Luoni et al. 2022). Moreover, datasets contribute significantly to optimizing shields based on source intensity and duration, allowing for efficacy in scenarios with varying radiation exposure durations. The consideration of attenuation characteristics, informed by extensive datasets, further refines shield optimization by addressing specific material interactions with radiation types (Ferrari et al. 2023; Sawik, 2023). Collectively, these aspects underscore the intricate relationship between radiation characteristics and dataset-driven strategies in crafting effective and adaptive shield designs.

The literature review systematically dissects studies employing extensive datasets to refine computational modeling parameters, an important aspect of shield optimization. The utilization of datasets ensures accuracy and reliability in simulations. Table 5 navigates the terrain of computational modeling parameters, showcasing the integral role of extensive datasets in refining and advancing shield optimization methodologies. Monte Carlo simulation parameters benefit significantly from extensive datasets, contributing to their refinement and, consequently, enhancing the precision and accuracy of simulations in shield optimization. Neutron and photon transport models find robust development and refinement through datasets, ensuring their computational representations align with the intricacies of shield optimization. The validation and benchmarking of computational models, important for ensuring their efficacy in predicting shield performance, are streamlined through the meticulous utilization of datasets. Computational resource requirements, a critical consideration, find guidance from datasets, optimizing the balance between model complexity and available resources.

Table 4 Radiation Characteristics and Dataset-Driven Optimization

Radiation Characteristic	Dataset-Driven Optimization in Shield Design	Source
Energy Levels and Spectra	Utilizing datasets with information on energy levels and spectra guides the optimization of shields for specific radiation sources	Qi et al. 2022
Interaction Cross-Sections	Comprehensive datasets on interaction cross-sections inform optimization strategies for shielding against specific radiation types	Bangalia 2024
Source Intensity and Duration	Datasets contribute to optimizing shields based on source intensity and duration, ensuring efficacy in scenarios with varying radiation exposure durations	Sadhu et al. 2024
Attenuation Characteristics	Extensive datasets inform the optimization of shields by considering the attenuation characteristics of materials against specific radiation types	El-Sawy and Sarwat 2024

Table 5 Computational Modeling Parameters and Dataset Refinement

Computational Parameter	Refinement through Utilization of Extensive Datasets	Source
Monte Carlo Simulation Parameters	Extensive datasets contribute to refining Monte Carlo simulation parameters, enhancing the accuracy of simulations in shield optimization	El-Sawy and Sarwat 2024
Neutron and Photon Transport Models	Datasets inform the development and refinement of neutron and photon transport models, ensuring robust computational representations for shield optimization	Chuong et al. 2024
Validation and Benchmarking Parameters	Utilizing datasets aids in the validation and benchmarking of computational models, ensuring their efficacy in predicting shield performance	Demircioğlu et al. 2024
Computational Resource Requirements	Datasets guide considerations of computational resource requirements, optimizing the balance between model complexity and available resources	Nazar et al. 2023
Sensitivity Analysis Parameters	Comprehensive datasets contribute to sensitivity analyses, refining parameters to assess the impact of uncertainties on the accuracy of computational models	Sayyed et al. 2024

Furthermore, comprehensive datasets play a pivotal role in sensitivity analyses, refining parameters to assess the impact of uncertainties on the accuracy of computational models (El-Sawy and Sarwat 2024; Bangalia 2024; Sadhu et al., (2024); ; ; ; ; ; ; ; ; ; . This comprehensive overview underscores the symbiotic relationship between computational modeling parameters and the rich understanding derived from extensive datasets in the realm of shield optimization.

10 Algorithms for data-driven optimization

ML algorithms emerge as pivotal tools in data-driven optimization, showcasing their prowess in discerning intricate patterns within radiation data. This facet enables the optimization of precision in material and design choices, a critical dimension in shielding efficacy. Studies underscore the efficacy of algorithms such as support vector machines, random forests, and neural networks in handling complex datasets, extracting meaningful patterns, and optimizing shielding strategies. The systematic analysis reveals a paradigm shift towards the integration of ML as a cornerstone in the evolution of data-driven optimization, emphasizing adaptability and real-time responsiveness (Fig. 3).

The literature scrutinizes the role of genetic algorithms in optimizing shielding designs. These evolutionary algorithms draw inspiration from natural selection processes, evolving potential shielding solutions over successive generations. Genetic algorithms exhibit a capacity to explore a vast solution space, adapting shield designs based on fitness criteria derived from extensive datasets (Bagheri and Khalafi 2023). This systematic exploration showcases their utility in navigating complex design spaces, enhancing the adaptability of shielding strategies to diverse radiation scenarios.

The systematic literature review unfolds a diverse array of optimization metaheuristics tailored to data-driven shield design. Particle swarm optimization, simulated annealing,

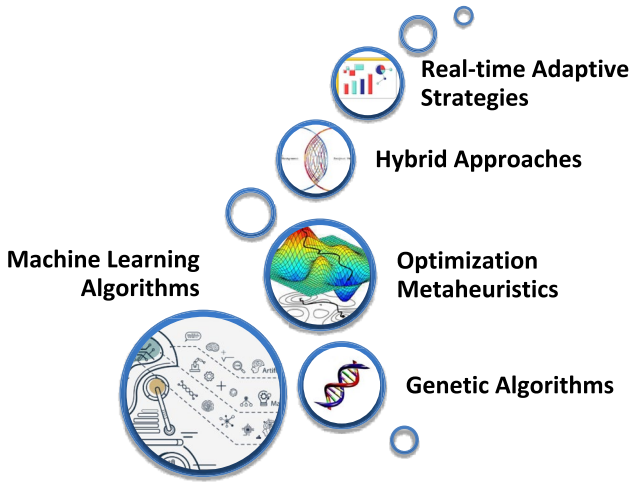


Fig. 3 Algorithms for Data-Driven Optimization

and ant colony optimization stand out as metaheuristic algorithms adept at navigating complex, multi-dimensional optimization spaces. These algorithms showcase their prowess in refining shielding strategies, dynamically adapting designs, and optimizing material choices. The literature accentuates their role in introducing adaptability and optimization precision to the forefront of shield design methodologies.

Studies explore the integration of diverse algorithms into hybrid frameworks, seeking to capitalize on the strengths of individual approaches. Hybridization of ML algorithms with genetic algorithms or optimization metaheuristics emerges as a promising avenue (Novello et al. 2024). This amalgamation harnesses the pattern recognition capabilities of ML alongside the exploration and adaptation prowess of genetic algorithms or metaheuristics. The systematic examination of hybrid approaches underscores their potential to offer a synergistic solution, leveraging the complementary strengths of different algorithms for enhanced optimization outcomes (Wang et al. 2023).

A noteworthy trend within the literature lies in the exploration of algorithms that facilitate real-time adaptive strategies. The integration of reinforcement learning algorithms, inspired by principles of reward-based learning, stands out. These algorithms exhibit a capacity to adapt shielding designs on-the-fly based on dynamic radiation scenarios (Pradhan & Koziel 2024). This real-time adaptability, highlighted in several studies, positions reinforcement learning as a key player in ushering in a new era of responsive and dynamically evolving shielding strategies.

11 Predictive modeling for shielding material properties

Reinforced composite materials certainly stand out lately, especially in the domain of shielding applications. As the interest for cutting edge shielding materials rises, designers are progressively going to predictive demonstrating to facilitate the advancement interaction and improve material properties.

11.1 Applications of predictive modeling

Predictive modeling arises as a key part in the design and refinement of reinforced composite materials customized for shielding applications, highlighting its critical job in different fields like aviation, thermal power, and clinical imaging. The strategic utilization of predictive models empowers researchers to finely calibrate material compositions and structures, achieving a harmonious equilibrium between weight, strength, and radiation attenuation properties.

In the aerospace industry, predictive modeling serves as a catalyst for innovation, steering the development of lightweight yet robust materials for aircraft shielding (Turczyn et al. 2020). Predictive models simulate the cumulative effects of high-energy particles on aerospace materials, accounting for long-term exposure to space radiation. Radiation transport simulations, degradation kinetics, and Monte Carlo methods help predict material behavior over extended periods. These models assess structural integrity, shielding effectiveness, and potential material breakdown caused by ionizing radiation. Real-world data from past space missions refine accuracy, enhancing reliability. Forecasting radiation-induced damage enables engineers to develop durable materials, ensuring spacecraft longevity and astronaut safety in extreme space environments. These models remain essential for mission planning, optimizing spacecraft design, and advancing protective measures against prolonged radiation exposure. Therefore, this sector demands materials capable of withstanding the unpredictable and dynamic conditions inherent in flight. Advanced modeling techniques furnish aerospace engineers with the means to simulate and analyze the behavior of composite materials across a spectrum of scenarios. Through these simulations, engineers can ensure that the selected materials not only meet safety principles but also offer optimal performance under the demanding conditions faced by aircraft.

Within the realm of nuclear energy, predictive modeling plays a central role in crafting shielding materials strategic for reactors and storage facilities (Chen and Yan 2023). The difficulties in this area are especially considerable, given the requirement for materials equipped for persevering through extraordinary radiation and thermal loads. Predictive modeling gives a vital device to expecting and grasping the way of behaving of reinforced composites in these harsh environments. Scientists can design shielding materials that add to the safety and life span of nuclear framework by recreating the material's reaction to extreme conditions.

The predictive demonstrating process requires a detailed comprehension of the perplexing interaction between the different constituents inside composite materials. This incorporates the matrix materials, reinforcement fibers, and any extra added substances, each impacting the general performance of the material. Predictive models, in this manner, become complex devices that record for the synergistic impacts of these parts, offering an extensive viewpoint on how the composite material will act under different environmental conditions.

11.2 Optimization of composite structures for radiation absorption

In the powerful domain of materials science, the optimization of composite designs for radiation absorption remains as an urgent and basic undertaking. This pursuit is propelled by the increasing demand for efficient radiation shielding materials that find applications across diverse industries, ranging from healthcare to aerospace and nuclear technology. Achieving optimal radiation absorption in composite structures entails a delicate balance between material composition, geometric design, and processing techniques as detailed in Fig. 4. In this intricate process, researchers employ a spectrum of methodologies, leveraging computational tools to model and simulate the intricate behavior of these structures under diverse radiation scenarios. Initially, simulations such as Monte Carlo methods and finite element modeling define optimization objectives and guide material selection for radiation absorption. Computational modeling helps manipulate material composition by predicting atomic interactions and optimizing nuclear number and density. It also refines geometrical designs through transport simulations and ray-tracing techniques. As materials are combined into composites, machine learning algorithms enhance optimization. Hence, computational modeling evaluates effectiveness and safety, ensuring maximum radiation attenuation

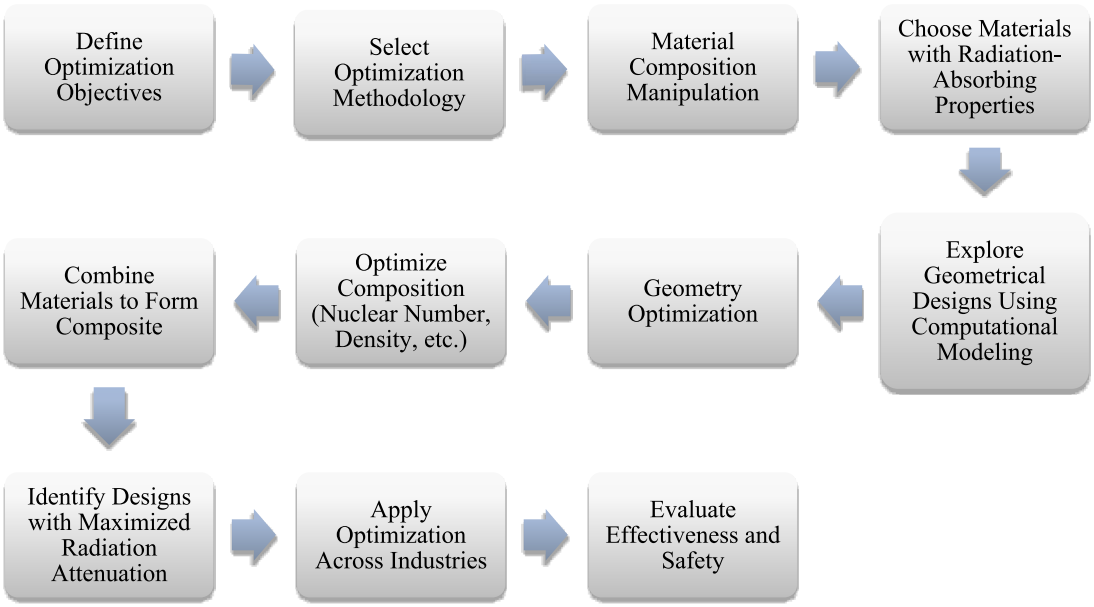


Fig. 4 Steps in the optimization of composite structures for radiation absorption

before industry application, reducing experimental costs, and accelerating the development of optimized shielding materials.

One of the foremost methodologies in optimizing composite structures for radiation absorption involves the manipulation of material composition. Researchers meticulously select and combine materials endowed with inherent radiation-absorbing properties, fostering the creation of composites with heightened attenuation characteristics (Jamwal et al. 2023). The optimization process is a detailed interplay of factors such as atomic number, density (Lin et al. 2023), and composition, aiming for a synergistic effect that ensures the material exhibits optimal radiation absorption capabilities (Mattos et al. 2023). In this regard, researchers seek to strike an equilibrium that maximizes the shielding efficiency of the composite structure.

Geometry optimization constitutes another indispensable facet of enhancing radiation absorption in composite structures. The spatial arrangement of reinforcing elements within the composite, coupled with considerations of the overall shape and density, plays a decisive role in determining the material's efficacy as a radiation shield. Computational modeling, a linchpin in this optimization journey, empowers researchers to explore an extensive array of geometrical designs. This exploration facilitates the identification of designs that promise to maximize radiation attenuation, ensuring the composite structure's effectiveness in mitigating the impact of various radiation sources.

The manipulation of composite composition and geometry through predictive modeling is fundamental in achieving multifaceted objectives. This approach extends its influence across diverse sectors, each with distinct requirements for radiation shielding materials. In the domain of medical services, where demonstrative imaging strategies like X-rays and CT scans are common, the optimization of composite designs becomes vital. Fitting materials to effectively absorb and lessen radiation is basic for patient safety as well as for accomplishing precise imaging results. The optimization cycle, directed by predictive demonstrating, considers the formation of shielding materials that find some kind of harmony between clearness in imaging and the safety of patients and medical care experts.

In the aerospace industry, where enormous and sun-oriented radiation present extraordinary difficulties, the streamlining of composite designs takes on an alternate aspect. The lightweight yet robust materials expected for air designed shielding require a fastidious equilibrium. Predictive demonstrating helps aeronautics designers in mimicking and anticipating the way of behaving of composite materials under the unique states of flight, guaranteeing that the materials fulfill safety guidelines as well as add to the general productivity and life span of air designed structures.

The nuclear area, set apart by its tough wellbeing necessities, depends vigorously on streamlined composite designs for radiation assimilation. Predictive modeling supports designing shielding materials for reactors and storage spaces, guaranteeing their strength under serious radiation and warm loads. The capacity to expect and reproduce the way of behaving of reinforced composites in such outrageous circumstances is instrumental in maintaining the wellbeing and functional respectability of nuclear framework.

As the field of materials science propels, the mix of predictive demonstrating in the optimization of composite designs for radiation absorption opens new boondocks. The collaboration between computational apparatuses and exploratory approval works with the formation of materials that not just meet the prerequisites of different ventures yet in addition add to more extensive progressions in wellbeing, productivity, and development. The continuous quest for enhancing composite designs for radiation absorption stays a demonstration of the constant responsibility of scientists and designers in pushing the limits of materials science to improve society (Lebedev et al. 2023).

Adaptive and real-time shielding systems address a change in outlook in the domain of radiation safety, offering a dynamic and responsive way to deal with relieve the effect of ionizing radiation. The interest for such frameworks has acquired unmistakable quality across different areas, including medical services, thermal power, and space investigation. The imperative for adaptive shielding systems arises from the inherently variable nature of radiation exposure scenarios. Conventional static shielding solutions, while effective, often fall short in addressing the unpredictability and dynamism of radiation sources. Adaptive shielding systems, in contrast, leverage real-time data and responsive mechanisms to dynamically adjust their shielding properties based on the prevailing radiation conditions.

In the medical care area, where clinical imaging methods are normal yet expose patients and medical care experts to ionizing radiation, the appearance of versatile shielding frameworks is especially huge. The capacity to momentarily adjust the degree of safety in light of the force and span of radiation exposure upgrades wellbeing as well as guarantees that clinical imaging stays exact and analytically successful.

The nuclear energy industry stands as another domain where the implementation of adaptive shielding systems holds immense promise. Reactor facilities, despite stringent safety protocols, are susceptible to fluctuations in radiation levels. The real-time monitoring capabilities of adaptive systems empower them to respond swiftly to changes in radiation intensity. Through effective optimization of shielding properties, these frameworks add to the upkeep of a safe functional environment while limiting the risk to work force and foundation.

Space investigation presents difficulties with respect to radiation exposure, given the shortfall of Earth's defensive air. Versatile and constant shielding frameworks become basic for guaranteeing the wellbeing of space travelers during expanded missions. The eccentric nature of inestimable radiation requires consistent observing and change of shielding boundaries. Responsive frameworks, fit for adjusting to changing radiation levels continuously, act as a pivotal defend for space travelers wandering into the universe.

Techniques utilized in the improvement of versatile and ongoing shielding frameworks rotate around the combination of cutting-edge sensors, data handling calculations and actuators (Deng et al. 2023). Sensors continuously monitor radiation levels, providing real-time data that serves as input for adaptive algorithms. These algorithms, often based on ML and artificial intelligence, analyze the incoming data to dynamically adjust the design of shielding materials or structures. Actuators, integral elements of these systems, execute the real-time adjustments, ensuring that the shielding properties align with the current radiation environment.

11.3 Integration of multi-physics and multi-scale approaches in radiation shielding

The combination multi-physics and multi-scale approaches in radiation shielding as shown in Fig. 5 addresses a momentous outskirt, reshaping our cognizance and calibrating the optimization of materials in the presence of ionizing radiation. The basic for coordinating multi-material science and multi-scale approaches emerges from the complicated nature of ionizing radiation interactions. Traditional shielding models, confined to specific physics domains or scales, often fail to capture the diverse and complex scenarios encountered in real-world applications. The integration of multi-physical science, addressing various radiation interactions at the same time, and multi-scale, representing material varieties at various length scales, offers a comprehensive and detailed worldview, hoisting the accuracy and predictive capacities of radiation shielding models. In a nutshell, multi-physics modeling accounts for interactions between radiation, thermal, mechanical, and electromagnetic properties, ensuring realistic assessments of shielding behavior. While, multi-scale approaches bridge atomic-level interactions with macroscopic material properties, refining structural integrity and radiation attenuation forecasts.

In the domain of room investigation, where astronomical radiation presents extraordinary difficulties, the coordination of multi-material science and multi-scale approaches is instrumental. Space apparatus shielding requests materials fit for enduring the ceaseless siege of high-energy particles (Li et al. 2022). Coordinating multi-physical science models empowers a far-reaching comprehension of the collaborations between different kinds of enormous radiation and shielding materials. Simultaneously, incorporating multi-scale approaches allows for a microscopic exploration of materials, considering crystal structures and atomic compositions to optimize shielding efficacy for long-duration space missions.

Beyond traditional contexts, the integration of these approaches finds relevance in homeland safety scenarios. The identification and moderation of radiological dangers request a refined comprehension of radiation interactions

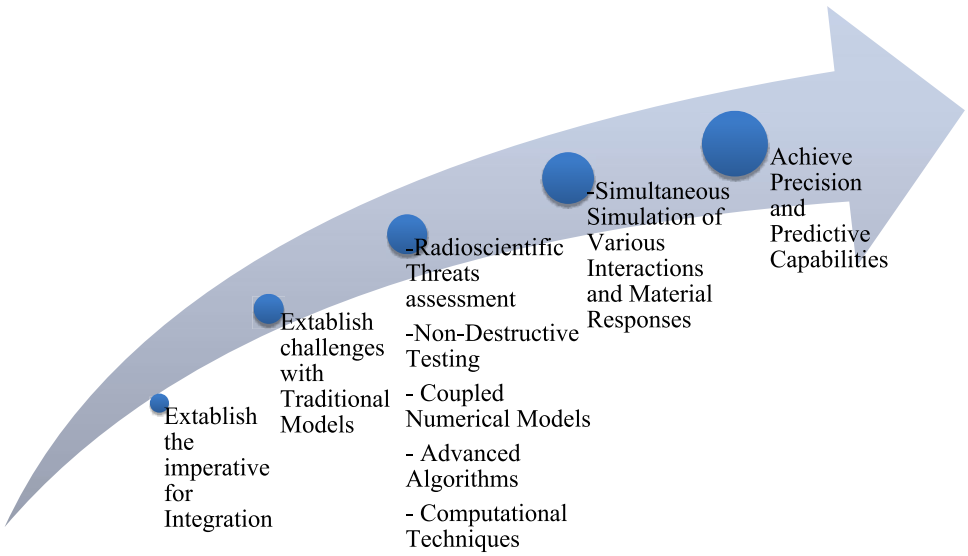


Fig. 5 Integration of Multi-Physics and Multi-Scale Approaches in radiation shielding

(Şakar et al. 2021). Multi-physical science models consider the recreation of different radiation sources and their interactions with shielding materials in metropolitan conditions. Integrating multi-scale considerations aids in optimizing the design of portable shielding systems, considering variations in material composition at the microscale to enhance their effectiveness in responding to radiological incidents (Dini et al. 2021).

In industrial radiography, where ionizing radiation is employed for non-destructive testing, the integration of multi-physics and multi-scale approaches contributes to enhanced safety and efficiency. Optimizing shielding materials for radiographic devices necessitates considering diverse radiation sources and their interactions within industrial settings. Multi-physics models facilitate the simultaneous consideration of these complex interactions, guiding the development of shielding materials tailored for specific industrial radiography applications (Wang et al. 2024). Incorporating multi-scale considerations further refines these models, accounting for variations in material composition and density at microscopic levels.

Scientifically, the integration of multi-physics and multi-scale approaches in radiation shielding involves the development of coupled numerical models (Bayat et al. 2021). These models intertwine different physics modules, allowing for the concurrent simulation of various radiation interactions and the exploration of material responses at multiple scales. Advanced algorithms and computational techniques are pivotal, ensuring the seamless interplay of diverse physics domains while maintaining computational efficiency and achieving accurate predictions.

12 Advanced material discovery

The application of ML transcends the realm of optimizing known materials for radiation shielding; it serves as a catalyst for the discovery of entirely novel shielding materials (Bilmez et al. 2022). This transformative aspect represents a paradigm shift in shield design, where ML algorithms harness the power of vast databases containing material properties and their correlations with shielding efficacy to identify promising candidates with superior radiation attenuation characteristics. This data-driven approach not only expedites the exploration of uncharted material territories but also fosters innovation, significantly broadening the spectrum of available options for radiation protection.

Traditionally, the search for effective shielding materials relied on a systematic exploration of known compositions, often guided by empirical rules and theoretical considerations. However, the landscape of potential materials is vast and intricate, making exhaustive exploration through conventional methods both time-consuming and resource intensive. ML steps in as a game-changer by leveraging its capacity to analyze large datasets efficiently, unveiling hidden patterns, and providing understanding that might elude traditional methodologies. The process begins with the compilation of extensive databases containing data on various material properties and their corresponding shielding performances. These databases encompass a diverse array of elements, compounds, and structures, creating a comprehensive repository that reflects the complexity of material interactions with ionizing radiation (Ruprecht et al. 2024). ML algorithms then

navigate through this expansive dataset, identifying correlations and patterns that indicate a material's efficacy in attenuating radiation.

Through effective discerning of these intricate relationships, ML algorithms can predict the potential shielding effectiveness of materials that may not have been previously considered. This predictive capability accelerates the screening process, allowing researchers to focus their attention on materials that exhibit promising characteristics in silico. This data-driven exploration expedites the identification of novel shielding candidates with superior radiation attenuation properties, thereby fostering innovation in shield design.

Furthermore, the ML-driven discovery of new shielding materials contributes to the expansion of the available options for radiation protection. Traditional materials may possess inherent limitations or trade-offs between key attributes such as weight, cost, and shielding efficacy (Saraswat et al. 2024). The introduction of novel materials, identified through ML methodologies, introduces fresh perspectives and possibilities. These materials may offer a more favorable combination of properties, addressing specific challenges in diverse applications, from nuclear reactors to medical facilities.

The innovative potential of ML in material discovery extends beyond incremental improvements, presenting opportunities for breakthroughs in radiation shielding. Materials with unconventional compositions or structures that may be neglected by regular methodologies can surface as feasible up-and-comers through the scientific ability of ML calculations (Juan et al. 2021). This widens the extent of accessible choices as well as presents a degree of flexibility and versatility significant for addressing developing difficulties in radiation safety.

13 Machine learning models for predicting shielding efficacy

Within the domain of predictive modeling, classification models emerge as prominent tools for assessing the shielding efficacy of materials against distinct radiation types. Studies showcase the efficacy of models such as support vector machines and random forests in classifying materials based on their ability to attenuate gamma rays, neutrons, or other ionizing radiation sources (Mathew et al. 2023). These models leverage labeled datasets, where materials are categorized according to their shielding efficiency, enabling the algorithms to discern patterns and establish predictive relationships.

Regression models, an integral facet of ML applications, play a pivotal role in predicting quantitative measures of shielding efficacy. These models are adept at estimating parameters such as half-value layer (HVL) and linear attenuation coefficient, providing detailed understanding into the material's ability to attenuate radiation as captured in Fig. 6. Researchers harness algorithms like linear regression and ensemble methods to establish predictive relationships between material properties and shielding effectiveness, offering valuable tools for fine-tuning material compositions.

Deep learning (DL) architectures, particularly neural networks, emerge as sophisticated mechanisms in predictive modeling for shielding efficacy (Jaha et al. 2022). These models showcase a capacity to discern intricate patterns within extensive datasets, allowing for a comprehensive understanding of the complex relationships between material properties and radiation attenuation. Convolutional neural networks (CNNs) prove effective in image-based predictions Krichen (2023), while recurrent neural networks (RNNs) showcase prowess in capturing

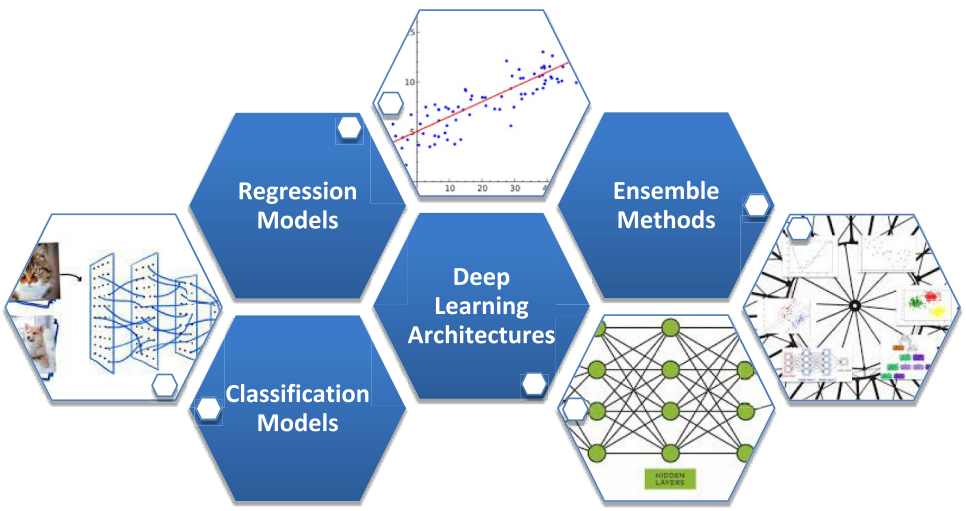


Fig. 6 Machine Learning Models for Predicting Shielding Efficacy

temporal dependencies, expanding the repertoire of predictive capabilities in the realm of shield design.

Ensemble methods, amalgamating the strengths of multiple models, stand out as robust tools in predictive modeling. Bagging and boosting techniques, exemplified by random forests and AdaBoost, respectively, showcase efficacy in enhancing predictive accuracy by combining the outputs of diverse ML models. This ensemble approach mitigates the risk of overfitting and enhances the generalizability of predictive models, contributing to a more reliable and adaptable framework for assessing shielding efficacy.

Literature underscores the significance of feature importance analysis in elucidating the key contributors to shielding efficacy predictions. ML models, particularly ensemble methods, offer understanding into the relative importance of various material properties in influencing the attenuation of ionizing radiation (Obrador and Montoro 2023). Feature importance analyses guide researchers in identifying critical factors and optimizing material compositions for enhanced shielding effectiveness, thereby fostering a data-driven approach in shield design. ML models prove versatile in handling the inherent complexity of shielding materials. The literature reflects an increasing focus on incorporating a multitude of material properties into predictive models, ranging from density and atomic number to specific structural features. This comprehensive approach acknowledges the synergistic effects of diverse constituents within materials, offering a detailed understanding of how complex compositions influence shielding efficacy.

14 Training datasets and validation strategies

The systematic exploration of literature unveils a meticulous approach to the construction and validation of training datasets in the realm of predicting shielding efficacy using ML models. This section covers the detailed methodologies and validation strategies, offering a glimpse into the systematic processes that underpin the development of robust predictive frameworks for shield design.

14.1 Construction of training datasets

Researchers exhibit a keen awareness of the pivotal role played by training datasets in enhancing the predictive accuracy of ML models for shielding efficacy. Table 6 synthesizes key aspects associated with the construction of training datasets evident in the literature.

The construction of training datasets involves a comprehensive consideration of material properties, radiation types, and shielding effectiveness labels. The inclusion of diverse material properties ensures a holistic representation of the shielding material, while the consideration of various radiation types caters to the adaptability of predictive models across different scenarios. Labeled datasets, classifying materials based on their shielding effectiveness, form the backbone of supervised learning, facilitating the training process (Narayanan et al. 2023; Bozzato et al. 2024). The incorporation of complex material models acknowledges the intricacies of shielding interactions, contributing to a more detailed and accurate predictive framework.

Table 6 Key Aspects in the Construction of Training Datasets

Aspect	Description	Source
Material Properties	Inclusion of diverse material properties such as density, atomic number, and composition	Luoni et al. 2022
Radiation Types	Consideration of various ionizing radiation types, including gamma rays, neutrons, etc	DeWitt & Benton 2024
Shielding Effectiveness Labels	Labeled datasets categorizing materials based on their shielding effectiveness	Fidan et al. 2024
Complexity of Material Models	Incorporation of complex material models to capture detailed interactions in predictions	Bozzato et al. 2024
Ensemble Learning for Feature Selection	Application of ensemble-based techniques (e.g., Random Forest, Gradient Boosting) to enhance feature selection by identifying the most influential parameters	Mathew et al. 2023
Model Improvement via Ensemble Learning	Utilization of ensemble methods to combine predictions from multiple models, reducing overfitting and improving generalization for shielding efficacy estimation	Pathan et al. 2024

14.2 Validation strategies

Validation strategies serve as critical elements in assessing the generalize ability and reliability of ML models for shielding efficacy prediction. Table 7 encapsulates key considerations and strategies evident in the literature.

Cross-validation techniques assume a strategic part in assessing the power of AI models, guaranteeing their presentation across different datasets. Outside approval against datasets not utilized in the preparation cycle gives experiences into the model's pertinence to concealed situations, improving its generalizability (Chaudhuri et al. 2024; Ali et al. 2024; Jang et al. 2020). Awareness examinations add to figuring out the effect of varieties in input boundaries on expectations, improving the dependability of the model under various circumstances. Similar evaluations against benchmark models or physical science-based reproductions act as benchmarks for approval, offering a reference point for the exactness and viability of the predictive structure.

15 Comparative analysis with traditional predictive models

The systematic exploration of literature reveals a compelling landscape wherein ML models for predicting shielding efficacy undergo a rigorous comparative analysis with traditional predictive models. This comprehensive review delves into the detailed methodologies and key findings derived from the juxtaposition of ML and traditional approaches, offering valuable understanding into the evolution and efficacy of predictive frameworks in the realm of shield design.

Table 7 Key Validation Strategies in the Literature

Strategy	Description	Source
Cross-Validation	Utilization of cross-validation techniques to assess model performance across diverse datasets	Killeen et al. 2024
External Validation	Validation against external datasets to evaluate the model's applicability beyond training data	Jang et al. 2020
Sensitivity Analysis	Rigorous sensitivity analyses to assess the impact of variations in input parameters on predictions	Chaudhuri et al. 2024
Comparative Assessments	Comparative assessments against benchmark models or physics-based simulations for validation	Ali et al. 2024

Table 8 Comparative Analysis Methodologies and Key Findings

Aspect	Description	Source
Methodologies		
Performance Metrics	Utilization of comprehensive performance metrics (e.g., accuracy, precision, recall, F1 score) for quantitative assessment of model effectiveness	Demircioğlu et al. 2024
Cross-Validation	Implementation of cross-validation techniques to ensure robust evaluation across diverse datasets and minimize overfitting concerns	Pathan et al. 2024
Sensitivity to Dataset Size	Examination of model sensitivity to dataset size variations, elucidating the impact on performance and generalizability	Amin et al. 2022
Feature Importance Analysis	Behavioring feature importance analyses to discern the key contributors to predictions in both ML and traditional models	Yin & Bingi, 2023
Statistical Significance Testing	Application of statistical significance testing to ascertain whether observed performance differences between models are statistically significant	Allen, 2023
Key Findings		
Enhanced Predictive Accuracy	ML models, particularly ensemble methods and DL architectures, often exhibit superior predictive accuracy compared to traditional models	Valizadeh et al. 2024
Adaptability to Complex Data	ML models demonstrate enhanced adaptability to complex datasets with diverse material properties and shielding scenarios	Wang et al. 2023
Sensitivity to Feature Changes	ML models showcase heightened sensitivity to changes in input features, enabling more detailed predictions in response to varying radiation environments	Lauwers et al. 2024
Interpretability Challenges	Traditional models, such as linear regression, may offer greater interpretability, facilitating a clearer understanding of the relationships between input parameters and predictions	Lee et al. 2023

15.1 Comparative analysis methodologies

The literature showcases a meticulous approach to comparative analyses, employing a variety of methodologies to assess the performance of ML models against traditional predictive models. Table 8 synthesizes key methodologies and considerations evident in the comparative analyses.

Comparative analyses employ a repertoire of performance metrics, ensuring a detailed understanding of the strengths and weaknesses of both ML and traditional predictive models. The adoption of cross-validation techniques contributes to a robust assessment, validating the models across diverse datasets and minimizing concerns related to overfitting. Evaluating sensitivity to dataset size provides understanding into the scalability and generalizability of models. Feature importance analyses elucidate the key contributors to predictions in both approaches, shedding light on the factors influencing shielding effectiveness. Statistical significance testing adds a layer of rigor, determining whether observed performance differences between models are statistically meaningful (Allen et al. 2023; Yin and Bingi 2023).

The literature unfolds a series of key findings derived from comparative analyses, elucidating the relative advantages of ML models in predicting shielding efficacy. Comparative analyses consistently highlight the enhanced predictive accuracy of ML models, especially in scenarios involving diverse material properties and complex shielding scenarios. ML models exhibit heightened adaptability to intricate datasets, offering detailed predictions that capture the synergistic effects of various material characteristics. Sensitivity to feature changes underscores the dynamic nature of ML models, enabling them to respond effectively to shifts in radiation environments (Lauwers et al. 2024).

While ML models excel in predictive accuracy, traditional models may retain an advantage in interpretability. Linear regression models, for instance, provide transparent understanding into the relationships between input parameters and predictions. However, this interpretability may come at the cost of capturing complex interactions present in shielding scenarios. Therefore, machine learning can significantly improve radiation shield design by optimizing material selection, reducing costs, and enhancing effectiveness. For example, ML models have identified novel composite materials with superior shielding properties, reducing reliance on expensive lead-based shields. In aerospace, ML-driven simulations have optimized lightweight shielding configurations for spacecraft, minimizing weight while maintaining radiation protection. Additionally, predictive models have accelerated testing, cutting down material development time and costs. In nuclear facilities, ML has refined shielding placement, reducing material waste while maximizing safety. These advancements demonstrate how ML enhances efficiency, cost-effectiveness, and performance in radiation shielding applications across various industries.

16 Scientific advancements in machine learning-assisted shield design

16.1 Data collection and preprocessing

The systematic exploration of the literature reveals a detailed landscape of scientific advancements in ML-assisted shield design, specifically focusing on data collection and preprocessing. In recent years, scientific advancements in data collection for ML-assisted shield design have witnessed a paradigm shift towards the acquisition of diverse and extensive datasets. Table 9 encapsulates key aspects evident in these methodologies.

Recent studies showcase a move towards comprehensive datasets derived from simulation-based approaches and experimental measurements, combining the strengths

Table 9 Data Collection Methodologies

Aspect	Description	Source
Simulation-Based Datasets	Utilization of sophisticated simulation tools to generate datasets emulating real-world radiation scenarios, providing a controlled and diverse training environment	Shi et al. 2023a, b
Experimental Measurements	Incorporation of experimental measurements from laboratory setups and real-world scenarios, ensuring the inclusion of actual radiation interactions for enhanced model generalizability	Lu et al. 2023
Comprehensive Material Properties	Integration of datasets encompassing a wide array of material properties, including density, composition, and thermal and mechanical characteristics, offering a holistic representation of shielding materials	Chaudhuri et al. 2024
Multi-Physics Data Integration	Exploration of multi-physics simulations that incorporate diverse radiation types, energy levels, and material behaviors, enriching datasets with comprehensive information	Riva et al. 2024
Real-Time Monitoring	Implementation of real-time monitoring systems to capture dynamic radiation scenarios, providing continuously updated datasets for adaptive ML models	Kawamura et al. 2024

of controlled simulations with real-world complexity. The inclusion of diverse material properties in datasets ensures a holistic representation of shielding materials, fostering adaptability in ML models. Furthermore, the integration of multi-physics simulations and real-time monitoring enhances the richness of datasets, enabling the exploration of dynamic radiation scenarios (Kawamura et al. 2024).

17 Data preprocessing techniques

The evolution of data preprocessing techniques in ML-assisted shield design is characterized by a refinement of methodologies to enhance the quality and relevance of input data. Table 10 synthesizes key aspects evident in these preprocessing techniques.

Late examinations grandstand an expanded accentuation on sound decrease strategies to upgrade the nature of the data, a pivotal move toward guaranteeing the precision and unwavering quality of AI models. Highlight scaling techniques add to the normalization of info highlights, addressing variations in scales and working with a more uniform effect during model preparation. Exception identification calculations assume a significant part in recognizing and correcting oddities, reinforcing the strength of AI models. Dimensionality decrease strategies are utilized to ease the difficulties presented by high-layered datasets, working on model effectiveness and attenuation computational requests. Moreover, dealing with worldly perspectives in datasets is progressively focused on, especially in situations including continuous checking, guaranteeing the significance of time-touchy information.

Recent studies underscore the integration of diverse ML frameworks tailored for shield design (Schütz et al. 2023). The literature points to the utilization of ensemble methods, DL architectures, and hybrid models that amalgamate the strengths of different algorithms. This integration is indicative of a holistic approach to ML, leveraging various frameworks to extract intricate patterns and optimize shielding strategies effectively.

18 Choice of machine learning algorithms

The systematic exploration of the literature unravels a compelling narrative surrounding scientific advancements in ML-assisted shield design, specifically delving into the critical domain of the choice of ML algorithms. This comprehensive review navigates through the rich tapestry of methodologies adopted in recent studies, shedding light on the evolution of algorithmic selections that underpin the integration of ML in the design and optimization of radiation shields (Sreenivas et al. 2023).

18.1 Ensemble methods

Ensemble methods arise as a foundation in ongoing strategic progressions, utilizing the force of numerous AI models to improve predictive precision and heartiness (Table 12). The combination of different calculations, for example,

Table 10 Data Preprocessing Techniques

Aspect	Description	Source
Noise Reduction	Application of noise reduction techniques to filter out irrelevant or erroneous data, ensuring the accuracy of input features for ML models	Cui 2024
Outlier Detection	Implementation of outlier detection algorithms to identify and address anomalies in the dataset, enhancing the robustness and reliability of ML models	Chen et al., 2024
Temporal Data Handling	Integration of methodologies for handling temporal aspects in datasets, particularly in scenarios involving real-time monitoring, ensuring the relevance of time-sensitive data	Schütz et al. 2023

Table 12 Ensemble Methods in Machine Learning-Assisted Shield Design

Aspect	Description	Source
Improved Predictive Accuracy	Ensemble methods, amalgamating predictions from diverse algorithms, exhibit enhanced accuracy compared to individual models, ensuring more reliable predictions	Li et al. 2023
Robustness to Noisy Data	The combination of multiple models provides inherent robustness to noisy or outlier-ridden datasets, contributing to the stability of predictions in diverse radiation scenarios	Zhang et al. 2022a, b
Handling Complex Interactions	Ensemble methods excel in capturing intricate patterns and complex interactions within radiation data, offering a more detailed understanding for optimal shield design	Shuryak et al. 2023
Mitigation of Overfitting	The ensemble approach mitigates overfitting concerns, striking a balance between model complexity and generalizability for improved performance on unseen data	Hu et al. 2024

Arbitrary Timberlands and Angle Helping, is obvious in late examinations.

The incorporation of ensemble methods represents a strategic move towards achieving heightened predictive accuracy and robustness in ML-assisted shield design. These methods, by leveraging the diversity of algorithms, showcase an enhanced ability to navigate complex radiation interactions and adapt to diverse shielding scenarios (Li et al. 2023; Hu et al. 2024).

18.2 Deep learning architectures:

Scientific advancements also manifest in the utilization of DL architectures, particularly convolutional neural networks (CNNs) and recurrent neural networks (RNNs), offering sophisticated frameworks capable of extracting intricate patterns from radiation data. Table 13 synthesizes key aspects evident in the integration of DL architectures.

The integration of DL architectures means a jump forward in the capacity to extricate complex examples from high-layered radiation datasets. These architectures, with their feature extraction capabilities and adaptability to spatial and temporal considerations, offer a detailed approach to shield design, particularly in scenarios where intricate patterns play a pivotal role (Salehi et al. 2023).

18.3 Hybrid models

Recent scientific advancements also highlight the emergence of hybrid models, integration the qualities of both conventional AI calculations and profound learning designs. The incorporation of these models presents an all-encompassing methodology, benefiting from the interpretability of traditional models and the example acknowledgment capacities of profound learning. The integration of hybrid models underscores a strategic approach, offering a synergy between the interpretability of traditional models and the advanced pattern recognition capabilities of DL architectures. These models provide a flexible and adaptive framework, catering to the diverse demands of radiation shield design.

19 Validation strategies and model interpretability

The systematic examination of the literature reveals a profound landscape of scientific advancements in ML-assisted shield design, with a specific focus on validation strategies and model interpretability. Validation is a critical facet in ensuring the reliability and generalizability of ML models for shield design (Huang et al. 2022). Recent studies showcase a detailed approach to validation strategies as shown in Fig. 7, emphasizing not only traditional metrics but also exploring novel techniques to enhance the robustness of models. The validation metrics carried out in machine learning for radiation shield design rely on mean squared error (MSE) and R-squared (R^2) to evaluate accuracy by comparing shielding effectiveness between predictions and measurements. However, these metrics may not capture generalization across different conditions. Novel validation strategies, including adversarial testing, domain adaptation, and uncertainty quantification, enhance robustness by evaluating performance under diverse and unseen scenarios. Multi-hospital validation and cross-domain testing ensure generalizability across varying radiation environments.

Cross-validation techniques stand out as a foundation in ongoing systemic advancements. Utilizing methods, for example, k-fold cross-validation guarantees that models are thoroughly tried across different subsets of the dataset, mitigating concerns connected with overfitting and improving

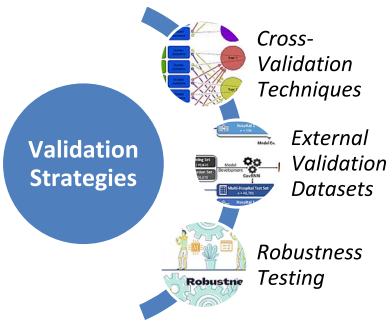


Fig. 7 Validation Strategies

Table 13 Deep Learning Architectures in Machine Learning-Assisted Shield Design

Aspect	Description	Source
Feature Extraction Capabilities	DL architectures excel in automatic feature extraction, allowing for the identification of complex patterns and relationships within radiation datasets	Dai et al. 2023a, b
Complex Pattern Recognition	DL models showcase unparalleled capabilities in recognizing intricate patterns within radiation data, providing a foundation for precise shield design strategies	Di et al. 2023
Transfer Learning Opportunities	The potential for transfer learning allows for the utilization of pre-trained models on related tasks, streamlining the training process and enhancing the efficiency of DL applications	Salehi et al. 2023

the model's generalizability (Tougui et al. 2021). Consolidating outer approval datasets is another vital pattern. Researchers increasingly recognize the importance of validating models on diverse datasets, sourced independently from the training data. This approach ensures that models exhibit efficacy across varied radiation scenarios contributing to their reliability in real-world applications. Robustness testing emerges as a important component of validation strategies.

Deciphering the choices made by AI models is basic, particularly in settings where the results of those choices influence wellbeing and viability. Late progressions center around improving model interpretability, guaranteeing that the reasoning behind shielding proposals is intelligible to area designers and end-clients.

Feature importance analysis remains a prevalent method to interpret ML models. Researchers employ techniques such as permutation importance or SHapley Additive exDesignations (SHAP) to discern the contribution of each input feature to the model's predictions. This analysis aids in identifying critical factors influencing radiation shielding recommendations. *LIME (Local Interpretable Model-agnostic Exdesignations)* represents a noteworthy advancement in enhancing model interpretability. This technique generates locally faithful exdesignations for individual predictions, providing insight into how specific instances influence the model's choices as described in Fig. 8. SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) enhance the interpretability of machine learning models in radiation shield design by explaining how input features influence predictions. SHAP assigns each feature a contribution score based on cooperative game theory, providing a global and local understanding of model behavior. LIME perturbs input data and trains an interpretable model to approximate the complex model's decisions locally. These techniques help identify critical shielding parameters, ensuring transparency and trustworthiness in ML-driven designs. LIME bridges the gap between complex ML models and human interpretability, particularly important in applications with high stakes, such as radiation shield design. Model-agnostic approaches gain prominence as they offer interpretability across various ML architectures (Li et al. 2023). Techniques like SHAP and LIME are not tied to specific models, allowing their application to diverse algorithms. This flexibility ensures that interpretability is not sacrificed, even when using complex models like DL architectures.

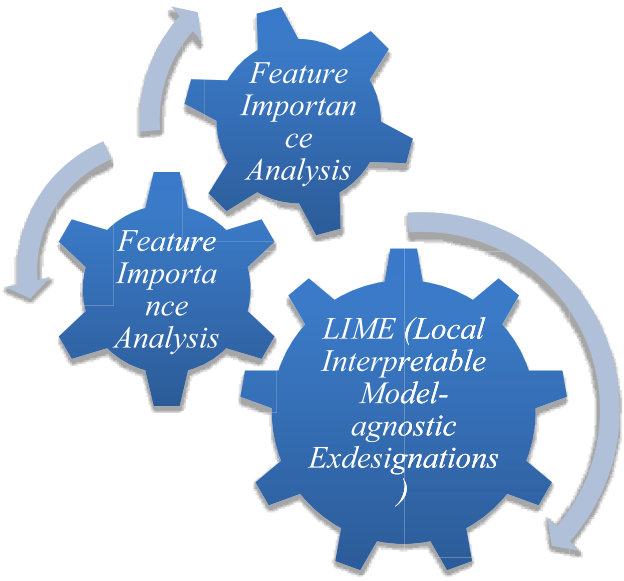


Fig. 8 Model Interpretability

20 Challenges, future directions and limitations

20.1 Limited availability of high-quality training data

ML has emerged as a valuable tool in optimizing shielding materials for radiation protection. However, the advancement of ML-assisted shield design encounters a persistent hurdle: the limited availability of high-quality training data as captured in Fig. 9. The cornerstone of effective ML models lies in the quality and diversity of the training data. Many studies in the literature have highlighted challenges related to the scarcity of diverse datasets representing various shielding scenarios. The limitation in data quality, arising from incomplete or outdated data, compromises the robustness and generalizability of ML models, hindering their real-world applicability (Ahmed-Mahmoodi et al. 2023). Collaborative efforts among research institutions, industries, and regulatory bodies are strategic to curate high-quality, diverse

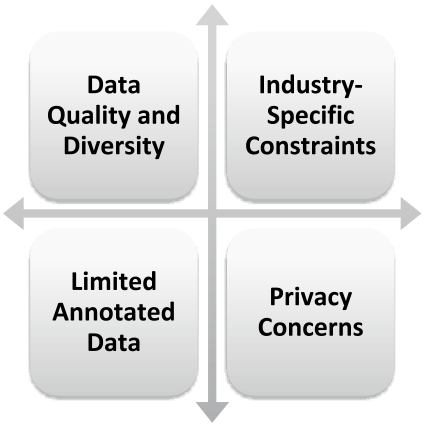


Fig. 9 Challenges of limited availability of high-quality training data

datasets. Establishing data-sharing platforms with principled protocols can encourage transparency and collective contributions to address the data quality challenge.

In certain industries, particularly those involving proprietary shielding materials or processes, access to high-quality training data is further constrained. The reluctance to share proprietary data poses a challenge in developing ML models that cater to the nuances of specific industrial applications, such as aerospace or nuclear energy. Industry collaboration is paramount. Establishing frameworks for responsible data sharing, where proprietary data is protected, can foster a collective approach to addressing industry-specific constraints. Encouraging pre-competitive collaboration ensures advancements without compromising individual interests.

Privacy considerations add another layer of complexity to the acquisition of training data. The sensitive nature of data, especially in healthcare or nuclear sectors, often limits the sharing of datasets due to privacy regulations. Striking a balance between data utility for model training and ensuring compliance with privacy principles remains a formidable challenge (Naito et al. 2020). Implementing ethical and legal frameworks that govern data sharing while shielding privacy is imperative. Engaging with stakeholders to establish transparent and ethical practices, including data anonymization and informed consent, can pave the way for responsible data utilization.

ML models often require annotated data for supervised learning, where each data point is labeled. The scarcity of annotated datasets for shield design poses a challenge in training models effectively. Annotating data for shielding materials involves domain expertise, and the shortage of annotated datasets hampers the development of accurate and reliable models. Active learning strategies and audience sourced annotation efforts can optimize the use of limited annotated data. Collaborative initiatives that involve domain experts in iterative data labeling processes can overcome the challenge of limited annotated datasets. Combining efforts across academia, industry, and regulatory bodies is vital to address the challenges of limited training data comprehensively. A holistic approach that integrates collaborative data-sharing practices, innovative data generation techniques, and ethical governance frameworks can propel ML-assisted shield design towards robust, responsible, and widely applicable solutions.

20.2 Transferability and generalization of ML models

The application of ML in shield design for radiation protection holds significant promise, yet it confronts a notable challenge: the transferability and generalization of ML models across diverse scenarios as captured in Fig. 10. The challenge often arises from the limited representation of diverse scenarios in the training datasets. If the training data inadequately covers the spectrum of radiation conditions and shielding materials, ML models may struggle to generalize to new and unforeseen situations (Ali et al. 2023). To address this challenge, future research should focus on creating more comprehensive and representative datasets. Collaborative efforts among researchers, industries, and regulatory bodies can contribute to the development of datasets that encapsulate a wider array of scenarios, ensuring that ML models are trained on a diverse and inclusive range of data.

Shield design spans various domains, each presenting unique challenges. ML models trained for a specific domain may struggle to generalize when applied to a different context. For instance, a model optimized for medical applications may not seamlessly transition to aerospace scenarios due to inherent differences in radiation sources and materials. A promising avenue is the development of domain-adaptive models that can effectively adapt to new domains. Integrating domain knowledge into the ML training process and exploring transfer learning techniques specific to shield design contexts can enhance the model's ability to generalize across diverse domains (Jiang et al. 2023; Xu et al. 2023).

The temporal and spatial dynamics of radiation scenarios pose challenges to model generalization. ML models trained on data from specific time periods or spatial locations may struggle to adapt to changes over time or in different geographical contexts. Machine learning models experience substantial difficulties when deployed to aerospace and nuclear energy sectors because of dissimilarities in data allocation along with environmental aspects and governing standards. Dataset representativeness is critical, as training

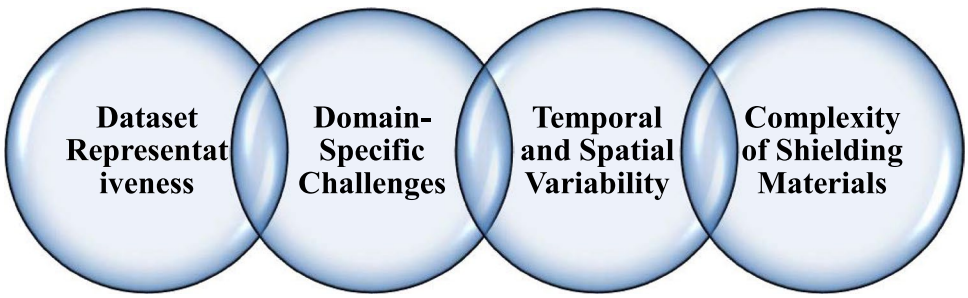


Fig. 10 Challenges of transferability and generalization of ML models

data may not cover extreme operational scenarios. Domain-specific challenges include highly specialized physics-based constraints and safety–critical requirements. Temporal and spatial variability affect model predictions, as radiation exposure and shielding effectiveness fluctuate across different conditions. Additionally, the complexity of shielding materials introduces non-linear interactions that traditional models may fail to capture. Robustness to temporal and spatial variability can be enhanced by incorporating dynamic features in the training data. Future research should explore the development of models that can adapt to changes over time and space, ensuring sustained effectiveness in evolving radiation environments. ML models may face challenges in generalizing across diverse shielding materials with varying complexities. The effectiveness of shielding materials can depend on intricate factors, including composition, structure, and environmental conditions, making it challenging for models to generalize without comprehensive understanding. Addressing the complexity of shielding materials requires advanced model architectures and innovative training techniques. Integrating physics-based simulations and incorporating material science principles into the ML training process can enhance the model's comprehension of complex shielding materials (Celozzi et al. 2023).

20.3 Interpretable machine learning for material data

The integration of ML into shield design processes brings forth a important consideration: the interpretability of ML models concerning material data (Fig. 11). ML models, especially complex ones like deep neural networks, are often regarded as black boxes, making it challenging to interpret their choice-making processes. This lack of interpretability is a significant hurdle, particularly in shield design where a clear understanding of how materials influence outcomes is critical. Researchers should explore the development of more interpretable ML models tailored to shield design. This involves leveraging simpler model architectures, such as choice trees or rule-based models, that provide explicit understanding into how material features contribute to shielding effectiveness.

In applications where safety is paramount, merely having accurate predictions is insufficient; understanding why a certain prediction was made is equally important. This is particularly relevant in shield design, where choices impact human safety and infrastructure integrity. Future research should prioritize the incorporation of explain ability into ML models. Techniques such as feature importance analysis, model-agnostic interpretability methods, and generating human-readable rules can enhance the interpretability of predictions, ensuring they align with domain expertise Gao et al. (2022). Shield design often involves adherence

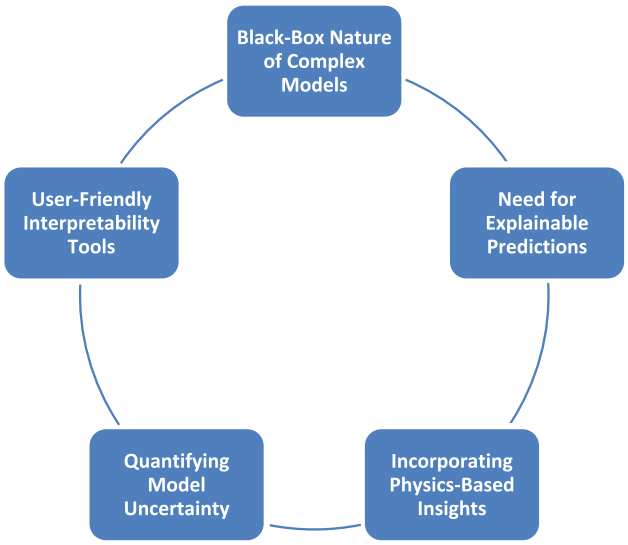


Fig. 11 Challenges relating to interpretable machine learning for material data

to physical principles governing radiation interactions with materials. Existing ML models may not explicitly incorporate these physics-based understanding, making it challenging to align predictions with the established understanding of material behaviors in radiation environments. The fusion of ML with physics-based models offers a promising direction. Integrating known physics principles into ML frameworks can enhance the interpretability of predictions, allowing for a more transparent alignment with the underlying material science and radiation physics.

Uncertainty in predictions is inherent, and understanding the uncertainty associated with ML model outputs is critical for informed choice-making. Shield design demands not only accurate predictions but also a quantification of the model's confidence or uncertainty in those predictions (Dai et al. 2023a, b). The development of models that provide uncertainty estimates alongside predictions is important. Bayesian approaches, ensemble methods, and probabilistic models can be explored to quantify uncertainties, providing valuable data for choice-makers in shield design scenarios. Practitioners and stakeholders in shield design may lack expertise in ML, making it strategic to provide user-friendly interpretability tools. Without accessible tools, the potential understanding from ML models may remain confined to the domain of ML experts.

In a nutshell, the three main obstacles preventing interpretability in shield design models through ML stem from their black-box nature along with the need for explainable predictions and challenges of including physics-based insights. Deep learning systems generally operate as opaque systems because they obscure the mechanics through which decisions occur. Safety applications require precise

measurement of model uncertainty because unsafe predictions in radiation shielding lead to destructive outcomes. Tools for interpretability should have friendly interfaces so users can convert ML knowledge into practical information. These barriers hinder trust, regulatory compliance, and informed decision-making, necessitating hybrid approaches that combine ML with domain expertise for reliable outcomes. The creation of user-friendly interpretability tools is imperative. Visualization techniques, interactive dashboards, and intuitive interfaces tailored to shield design professionals can bridge the gap, enabling non-experts to leverage the understanding generated by ML models effectively.

Future research needs to handle interpretability challenges in radiation shield design through mathematical approaches that combine ML with physical simulations to create models with scientific accuracy. The interpretability of AI systems can be enhanced through the use of Explainable AI technique SHAP and LIME to reveal important features that drive prediction results. Research teams should employ transfer learning as well as synthetic data generation to address data availability challenges using limited experimental datasets. Research institutions can enhance their access to various high-quality data through collaborative frameworks that enable data sharing among institutions. The implementation of uncertainty quantification methods improves the trust in ML models which results in better reliability for safety–critical applications.

20.4 Limitations

The research depends on publicly available databases and such resources may present some limitations to their real-life usage. The collected datasets from literature may lack comprehensive material and environmental coverage thus restricting universal applicability of discovered findings. Although the gathered data are valuable, it may not always match some real-life application where conditions can vary significantly. On the other hand, machine learning algorithms create difficulties when used for predicting purposes, this is because, most predictive models exist as unexplainable black boxes regarding their evaluation procedures. The absence of clarity about operating systems makes it challenging to validate outcomes particularly when industries like healthcare and aerospace need absolute comprehension of prediction reasoning. The study examines primarily three sectors which include healthcare and aerospace and nuclear energy and does not consider defense and environmental protection or similar crucial fields. Future research investigations into these industries would contribute to building an improved understanding of programable system applications. Because this is a review study, an absence of appropriate experimental studies produces uncertain real-world reliability of these findings. Therefore, an additional real-world testing and validation procedures must be performed to confirm the findings outside theoretical framework analysis.

21 Conclusions

The systematic review of research advancements in ML-assisted design of reinforced composite radiation shields has illuminated the transformative impact of ML on the field. Researchers have achieved remarkable milestones in optimizing composite structures for radiation absorption by delicately balancing material composition, geometric design, and processing techniques. The meticulous manipulation of composite composition, guided by predictive modeling, has led to the creation of materials with heightened radiation absorption capabilities. Geometry optimization, empowered by computational modeling, has played a decisive role in enhancing the efficacy of composite structures as radiation shields. These advancements find diverse applications in industries such as healthcare, aerospace, and the nuclear sector, where the demand for efficient radiation shielding materials continues to rise.

The integration of predictive modeling and ML techniques has revolutionized the optimization of composite structures for radiation absorption. The synergy between computational tools and experimental validation has paved the way for materials that not only meet specific industry requirements but also contribute to broader advancements in safety, efficiency, and innovation. In healthcare, the tailored optimization of composite structures ensures both patient safety and accurate diagnostic imaging. The aerospace industry benefits from lightweight, yet robust materials designed to withstand cosmic and solar radiation, contributing to the longevity and efficiency of aircraft structures. In the nuclear sector, where safety requirements are stringent, ML-assisted design ensures the resilience of shielding materials under extreme radiation and thermal loads, upholding the operational integrity of nuclear infrastructure.

As we conclude this systematic review, the implications for future research in the ML-assisted design of reinforced composite radiation shields are substantial. The field is poised for further exploration, with several key avenues deserving attention. Firstly, advancing interpretability in ML models for material data is critical. Ensuring that these models produce explainable predictions aligned with physics-based understanding will foster trust and facilitate broader adoption in the field. Secondly, addressing the challenges of limited availability of high-quality training data is pivotal. Collaborative efforts to create representative datasets, coupled with innovative data generation techniques, can propel the field forward. Thirdly, enhancing the transferability and generalization of ML models across diverse scenarios remains a pressing concern. Developing domain-adaptive

models and accounting for temporal and spatial variability will be instrumental in overcoming this challenge. Moreover, the quest for privacy-aware ML in shield design is an avenue that demands attention. Ethical considerations encompassing predisposition, decency, transparency, responsibility, and protection should be incorporated into the turn of events and organization of ML models. Laying out vigorous ethical systems, complying to lawful guidelines, and consolidating safety saving methods will add to responsible advancements in the field. The interdisciplinary nature of this field requires progressing joint effort among materials researchers, AI designers, and space subject matter experts. Continuous innovation in ML algorithms, data-driven approaches, and ethical considerations will shape the future landscape of reinforced composite shield design.

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Declarations

Competing interest The authors declare that there is no competing interest.

Research involving human and /or animals Not applicable.

Informed consent Not applicable.

References

Abulyazied DE, Issa SA, Alrowaily AW, Saudi HA, Zakaly HM, Ali ES (2023) Polylactic acid tungsten trioxide reinforced composites: a study of their thermal, optical, and gamma radiation attenuation performance. Radiat Phys Chem 205:110705. https://doi.org/10.1016/j.radphyschem.2022.110705

Abu-Saleem R, Abdelal N, Alsabbagh A, Al-Jarrah M, Al-Jawarneh F (2021) Radiation shielding of fiber reinforced polymer composites incorporating lead nanoparticles—an empirical approach. Polymers 13(21):3699. https://doi.org/10.3390/polym13213699

Aggarwal R, Sounderajah V, Martin G, Ting DS, Karthikesalingam A, King D, Ashrafian H, Darzi A (2021) Diagnostic accuracy of deep learning in medical imaging: a systematic review and meta-analysis. NPJ Digital Medicine 4(1):65. https://doi.org/10.1038/s41746-021-00438-z

Ahmed-Mahmoodi MM, Ahmmad SK, Ramadevudu G, Rammah YS, Laxman Naik J (2023) Structure, radiation shielding characteristics, bioactive studies and artificial intelligence density prediction of mixed alkali B2O3-SiO2-CaO-ZnO glasses. SILICON 15(18):7979–7993. https://doi.org/10.1007/s12633-023-02645-7

Al-Hassan M, Wang Z, Liu WB, Wang J, Zhigang Y, Khan M, Ali MM, Geldiyev R, Diaby M, Derradji M (2021) Thermal stability and gamma ray shielding properties of tungsten borides/epoxy micro-composites. Radiat Phys Chem 189:109769. https://doi.org/10.1016/j.radphyschem.2021.109769

Ali S, Ghatwary N, Jha D, Isik-Polat E, Polat G, Yang C, Li W, Galdran A, Ballester MÁG, Thambawita V, Hicks S (2024) Assessing generalisability of deep learning-based polyp detection and segmentation methods through a computer vision challenge. Sci Rep 14(1):2032. https://doi.org/10.1038/s41598-024-52063-x

Ali, H., Muthudoss, P., Chauhan, C., Kaliappan, I., Kumar, D., Paudel, A., & Ramasamy, G. (2023). Machine Learning-Enabled NIR Spectroscopy. Part 3: Hyperparameter by Design (HyD) Based ANN-MLP Optimization, Model Generalizability, and Model Transferability. *AAPS PharmSciTech*, *24*(8), 254. https://doi.org/10.1208/s12249-023-02697-3

Allen, G. I., Gan, L., & Zheng, L. (2023). Interpretable machine learning for discovery: Statistical challenges and opportunities. *Annual Review of Statistics and Its Application*, *11*. https://doi.org/10.1146/annurev-statistics-040120-030919

Alresheedi MT, Elsafi M, Aladadi YT, Abas AF, Ganam AB, Sayyed MI, Mahdi MA (2023) Assessment of silicone rubber/lead oxide composites enriched with Bi2O3, WO3, BaO, and SnO2 nanoparticles for radiation shielding applications. Polymers 15(9):2160. https://doi.org/10.3390/polym15092160

Al-Zaman MA, Nizam QMR (2022) Study on shielding effectiveness of a combined radiation shield for manned long termed interplanetary expeditions. J Space Saf Eng 9(1):83–89. https://doi.org/10.1016/j.jsse.2021.12.003

Ambrose, K. (2023). Conservatism versus sustainability: recognising the interconnectivity of ALARA and sustainability. *Radiation Protection Dosimetry*, *199*(8–9), 747–753. s10.1093/rpd/ncad075

Amin MN, Ahmad I, Iqbal M, Abbas A, Khan K, Faraz MI, Alabdullah AA, Ullah S (2022) Computational AI models for investigating the radiation shielding potential of high-density concrete. Materials 15(13):4573. https://doi.org/10.3390/ma15134573

Amin M, Elemam WE, Kandil MA, Samy M (2023) Effect of heavyweight recycled materials on radiation shielding and properties of high-strength concrete with CEM III. J Build Eng 79:107819. https://doi.org/10.1016/j.jobe.2023.107819

Ampanavos, S., Nourbakhsh, M., & Cheng, C. Y. (2021, July). Structural Design Recommendations in the Early Design Phase Using Machine Learning. In *International Conference on Computer-Aided Architectural Design Futures* (pp. 190–202). Singapore: Springer Singapore. https://doi.org/10.1007/978-981-19-1280-1_12

Athar MS, Fatima A, Singh SK (2023) Neutrinos and their interactions with matter. Prog Part Nucl Phys 129:104019. https://doi.org/10.1016/j.ppnp.2022.104019

Azman MN, Abualroos NJ, Yaacob KA, Zainon R (2023) Feasibility of nanomaterial tungsten carbide as lead-free nanomaterial-based radiation shielding. Radiat Phys Chem 202:110492. https://doi.org/10.1016/j.radphyschem.2022.110492

Bagheri S, Khalafi H (2023) SMR, 3D source term simulation for exact shielding design based on genetic algorithm. Ann Nucl Energy 191:109915. https://doi.org/10.1016/j.anucene.2023.109915

Baker SI, Kamboj S (2022) Applying ALARA Principles in the Design of New Radiological Facilities. Health Phys 122(3):452–462. https://doi.org/10.1097/HP.0000000000001515

Bangalia, S. (2024). Uncertainties and Biases in Neutrino-Nucleus Interactions at DUNE Far Detector: A Machine Learning Approach (No. CERN-STUDENTS-Note-2024–001).

Baroň I, Koktavý P, Trčka T, Rowberry M, Stemberk J, Balek J, Plan L, Melichar R, Diendorfer G, Macků R, Škarvada P (2022) Differentiating between artificial and natural sources of electromagnetic radiation at a seismogenic fault. Eng Geol 311:106912. https://doi.org/10.1016/j.enggeo.2022.106912

Bassam SA, Naseer KA, Mahmoud KA, Sangeeth CS, Sayyed MI, Alqahtani MS, El Shiekh E (2024) Examination of the ionizing radiation shielding behavior of the zinc boro-tellurite glasses doped with dysprosium oxide. Radiat Phys Chem 215:111359. https://doi.org/10.1016/j.radphyschem.2023.111359

Bayat M, Dong W, Thorborg J, To AC, Hattel JH (2021) A review of multi-scale and multi-physics simulations of metal additive manufacturing processes with focus on modeling strategies. Addit Manuf 47:102278. https://doi.org/10.1016/j.addma.2021.102278

Bhamare DK, Saikia P, Rathod MK, Rakshit D, Banerjee J (2021) A machine learning and deep learning based approach to predict the thermal performance of phase change material integrated building envelope. Build Environ 199:107927. https://doi.org/10.1016/j.buildenv.2021.107927

Bhosekar A, Ierapetritou M (2020) Modular design optimization using machine learning-based flexibility analysis. J Process Control 90:18–34. https://doi.org/10.1016/j.jprocont.2020.03.014

Bilmez B, Toker O, Alp S, Öz E, İçelli O (2022) A comparative study on applicability and efficiency of machine learning algorithms for modeling gamma-ray shielding behaviors. Nucl Eng Technol 54(1):310–317. https://doi.org/10.1016/j.net.2021.07.031

Boahen JK, Elsagheer Mohamed SA, Khalil AS, Hassan MA (2023a) Application of artificial intelligence techniques in modeling attenuation behavior of ionization radiation: a review. Radiation Detect Technol Methods 7(1):56–83. https://doi.org/10.1007/s41605-022-00368-8

Boahen JK, Khalil AS, Hassan MA, Mohamed SAE (2023b) EJUSTCO: Monte Carlo radiation transport code hybrid with ANN model for gamma-ray shielding simulation. Nucl Sci Tech 34(9):144. https://doi.org/10.1007/s41365-023-01297-x

Bosikov, I. I., Klyuev, R. V., Silaev, I. V., & Pilieva, D. E. (2023). Estimation of multistage hydraulic fracturing parameters using 4D simulation. *Gornye nauki i tekhnologii= Mining Science and Technology (Russia)*, *8*(2), 141–149. https://doi.org/10.17073/2500-0632-2023-01-97

Boukhris I, Kebaili I, Al-Buriahi MS, Sayyed MI (2021) Radiation shielding properties of tellurite-lead-tungsten glasses against gamma and beta radiations. J Non-Cryst Solids 551:120430. https://doi.org/10.1016/j.jnoncrysol.2020.120430

Bozzato, D., Dyrcz, P., Froeschl, R., Infantino, A., Lorenzon, T., Tisi, M., & Vincke, H. (2024). Advanced simulation techniques for Radiation Protection studies at the Large Hadron Collider. *Radiation Physics and Chemistry*, 111573. https://doi.org/10.1016/j.radphyschem.2024.111573

Cao Q, Gao Y, Pu J, Elshahawy AM, Guan C (2024) Materials and structural design for preferable Zn deposition behavior toward stable Zn anodes. SmartMat 5(1):e1194. https://doi.org/10.1002/smm2.1194

Celozzi, S., Araneo, R., Burghignoli, P., & Lovat, G. (2023). *Electromagnetic Shielding: Theory and Applications*. John Wiley & Sons.

Chao M, El Naqa I, Bakst RL, Lo YC, Peñagarícano JA (2022) Cluster model incorporating heterogeneous dose distribution of partial parotid irradiation for radiotherapy induced xerostomia prediction with machine learning methods. Acta Oncol 61(7):842–848. https://doi.org/10.1080/0284186X.2022.2073187

Chaudhuri SK, Li Q, Mandal KC, Hu J (2024) Deep learning-based classification of gamma photon interaction in room-temperature semiconductor radiation detectors. IEEE Access. https://doi.org/10.1109/ACCESS.2024.3354270

Chauhan, V., Hamada, N., Garnier-Laplace, J., Laurier, D., Beaton, D., Tollefsen, K. E., & Locke, P. A. (2022). Establishing a communication and engagement strategy to facilitate the adoption of the adverse outcome pathways in radiation research and regulation. *International journal of radiation biology*, *98*(12), 1714-1721. https://doi.org/10.1080/09553002.2022.2086716

Chen YQ, Yan BH (2023) The technology of shielding design for nuclear reactor: a review. Prog Nucl Energy 161:104741

Chen Y, Tian Y, Zhou Y, Fang D, Ding X, Sun J, Xue D (2020) Machine learning assisted multi-objective optimization for materials processing parameters: a case study in Mg alloy. J Alloy Compd 844:156159

Chen H, Li X, Feng Z, Wang L, Qin Y, Skibniewski MJ, Chen ZS, Liu Y (2023) Shield attitude prediction based on Bayesian-LGBM machine learning. Inf Sci 632:105–129

Cho YI, Shin SH, Kim JH (2020) Dose assessment and shielding analysis of the working environments for non-destructive testing. J Instrum 15(06):P06012

Chuong HD, Khang NHD, Linh NTT, Sang TT, Dat NT, Tam HD (2024) A comparative study on full-energy peak efficiency calibration methods for HPGe detector: Virtual point detector, curve fitting, and machine learning models. Nucl Instrum Methods Phys Res Sect A 1058:168941

Ciapessoni E, Cirio D, Pitto A (2023) A cost-benefit analysis framework for power system resilience enhancement based on optimization via simulation considering climate changes and cascading outages. Energies 16(13):5160

Cordella F, Cappelli M, Ciotti M, Claps G, De Leo V, Mazzotta C, Pacella D, Tamburrino A, Panza F (2024) Genetic algorithm for multilayer shield optimization with a custom parallel computing architecture. Euro Phys J plus 139(2):150

Cui E (2024) Reducing radiation dose in routine CT scans: an AI-driven approach with deep learning–based dual-energy CT reconstruction. Eur Radiol 34(1):26–27

Dai D, Cao L, Liu Y, Wang Y, Wu Z (2023a) Classification of high-altitude flying objects based on radiation characteristics with attention-convolutional neural network and gated recurrent unit network. Remote Sens 15(20):4985

Dai Z, Li P, Zhu M, Zhu H, Liu J, Zhai Y, Fan J (2023b) Dynamic prediction for attitude and position of shield machine in tunneling: A hybrid deep learning method considering dual attention. Adv Eng Inform 57:102032

Dastider AG, Saha S, Islam Sukanya M, Chakraborty R (2021) Comparative analysis among materials for passive shielding in a manned Mars mission. Astrophys Space Sci 366:1–10

Demircioğlu A, Bos D, Demircioğlu E, Qaadan S, Glasmachers T, Bruder O, Umutlu L, Nassenstein K (2024) Deep learning–based scan range optimization can reduce radiation exposure in coronary CT angiography. Eur Radiol 34(1):411–421

Deng Q, Jia H, An C, Wu S, Zhao S, Hu N (2023) Progress and prospective of electrochemical actuator materials. Compos A Appl Sci Manuf 165:107336

Dercle L, Henry T, Carré A, Paragios N, Deutsch E, Robert C (2021) Reinventing radiation therapy with machine learning and imaging bio-markers (radiomics): State-of-the-art, challenges and perspectives. Methods 188:44–60

DeWitt, J. M., & Benton, E. R. (2024). Secondary Proton Buildup in Space Radiation Shielding. *Life Sciences in Space Research*.

Di Y, Wang E, Huang T (2023) Identification method for microseismic, acoustic emission and electromagnetic radiation interference signals of rock burst based on deep neural networks. Int J Rock Mech Min Sci 170:105541

Dini D, Cognigni F, Passeri D, Scaramuzzo FA, Pasquali M, Rossi M (2021) Multiscale characterization of Li-ion batteries through the combined use of atomic force microscopy and X-ray microscopy and considerations for a correlative analysis of the reviewed data. J Electrochem Soc 168(12):126522

Dorise, A., Alonso, C., Subias, A., Travé-Massuyès, L., Baczkowski, L., & Vacher, F. (2021). Machine learning as an alternative to thresholding for space radiation high current event detection. In *2021 21th European Conference on Radiation and Its Effects on Components and Systems (RADECS)* (pp. 1–7). IEEE.

Elbaz K, Zhou A, Shen SL (2023) Deep reinforcement learning approach to optimize the driving performance of shield tunnelling machines. Tunn Undergr Space Technol 136:105104

El-Khatib AM, Doma AS, Badawi MS, Abu-Rayan AE, Aly NS, Alzahrani JS, Abbas MI (2020) Conductive natural and waste rubbers composites-loaded with lead powder as environmental flexible gamma radiation shielding material. Mater Res Express 7(10):105309

El-Khatib AM, Abbas MI, Mahmoud ME, Fayez-Hassan M, Dib MF, Khalil MH, El Aal AA (2023) Developing of lead/polyurethane micro/nano composite for nuclear shielding novel supplies: γ-spectroscopy and FLUKA simulation techniques. Polymers 15(22):4416

El-Mesady IA, Rammah YS, Hussein AE, El-Samman HM, El-Agawany FI, Elsad RA (2023) Synthesis, optical, mechanical characteristics, and gamma-ray shielding capacity of polyethylene-basalt mixture. Radiat Phys Chem 209:110974

Elsafi M, Al-Ghamdi H, Sayyed MI, Antar A, Almuqrin AH, Mahmoud KA, Cornish K, Shalaby TI, El-Khatib AM (2022) Optimizing the gamma-ray shielding behaviors for polypropylene using lead oxide: a detailed examination. J Market Res 19:1862–1872

El-Sawy AA, Sarwat E (2024) Comparative Study of Gamma-Ray Shielding Parameters for Different Epoxy Composites. Baghdad Sci J 21(2):0480–0480

Ferrari, M., Alía, R. G., Giles, T., Senajova, D., Pandini, S., Zenoni, A., & Calviani, M. (2023). 'Radiation to Materials' at CERN. *IEEE Transactions on Nuclear Science*.

Fetzer A, Anger M, Oleynik P, Praks J (2024) Total ionising dose multilayer shielding optimisation for nanosatellites on geostationary transfer orbit. Adv Space Res 73(1):831–845

Fidan M, Acikgoz A, Yılmaz D, Demircan G, Kalecik S, Aktas B, Isgor S (2024) Investigation of the structural, mechanical, radiation and neutron shielding properties of the TeO2-B2O3-Li2O-MoO3-CuO glass system. J Alloy Compd 976:172981

Ford E, Maneparambil K, Rajan S, Neithalath N (2021) Machine learning-based accelerated property prediction of two-phase materials using microstructural descriptors and finite element analysis. Comput Mater Sci 191:110328

Fragkia, V., Foged, I. W., & Pasold, A. (2021). Predictive information modeling: machine learning strategies for material uncertainty. *Technology| Architecture+ Design*, *5*(2), 163–176.

Galdino F, Silva J, Silva G, Pinheiro F, Rodrigues M, Sousa V, Barbosa H, Silva R, Carvalho V, Medeiros L, Silva J (2023) Measurement and analysis of non-ionizing radiation levels in residential environments emitted by Wi-Fi routers. IEEE Lat Am Trans 22(1):31–38

Gao Y, Zhu Y, Zhao Y (2022) Dealing with imbalanced data for interpretable defect prediction. Inf Softw Technol 151:107016

Guo K, Yang Z, Yu CH, Buehler MJ (2021) Artificial intelligence and machine learning in design of mechanical materials. Mater Horiz 8(4):1153–1172

Hamisu, A., Khiter, O., Al-Zhrani, S., Haridh, W. S. B., Al-Hadeethi, Y., Sayeed, M. I., & Tijani, S. A. (2023). The use of nanomaterial polymeric materials as ionizing radiation shields. *Radiation Physics and Chemistry*, 111448.

Han D, Kwon S (2021) Application of machine learning method of data-driven deep learning model to predict well production rate in the shale gas reservoirs. Energies 14(12):3629

Hassanpour, M., Hassanpour, M., Khandaker, M. U., Faruque, M. R. I., Alshahrani, B., & Osman, H. (2023). An alternative method for calculation of half-value layers without the knowledge of attenuation coefficient. *Applied Radiation and Isotopes*, 110910.

Hasti VR, Navarkar A, Gore JP (2021) A data-driven approach using machine learning for early detection of the lean blowout. Energy and AI 5:100099

Hemath M, Mavinkere Rangappa S, Kushvaha V, Dhakal HN, Siengchin S (2020) A comprehensive review on mechanical, electromagnetic radiation shielding, and thermal conductivity of fibers/inorganic fillers reinforced hybrid polymer composites. Polym Compos 41(10):3940–3965

Hu D, Wang Y, Ji G, Liu Y (2024) Using machine learning algorithms to predict the prognosis of advanced nasopharyngeal carcinoma after intensity-modulated radiotherapy. Curr Probl Cancer 48:101040

Huang H, Chang J, Zhang D, Zhang J, Wu H, Li G (2022) Machine learning-based automatic control of tunneling posture of shield machine. J Rock Mech Geotech Eng 14(4):1153–1164

Ihueze, C.C., Onwurah, U.O., Okafor, C.E., Obuka, N.S., Okpala, C.C., Okoli, N.C., Nwankwo, C.O. and Kingsley-Omoyibo, Q.A., (2023). Robust design and setting process and material parameters for electrical cable insulation. *The International Journal of Advanced Manufacturing Technology*, pp.1–18.

Iqbal, J., Howari, F. M., Mohamed, A. M. O., & Paleologos, E. K. (2021). Assessment of radiation pollution from nuclear power plants. In *Pollution Assessment for Sustainable Practices in Applied Sciences and Engineering* (pp. 1027–1053). Butterworth-Heinemann.

Jaha N, Islam GS, Kabir MF, Khandaker MU, Bhuian ASI (2022) Ionizing radiation shielding efficacy of common mortar and concrete used in Bangladeshi dwellings. Case Stud Construct Mater 17:e01547

Jamwal U, Singh D, Yadav KL (2023) An experiment to observe the effect of gamma radiation on microwave absorption properties of Li-substituted nickel ferrite. J Mater Sci 34(22):1643

Jang M, Kim M, Bae SJ, Lee SH, Koh JM, Kim N (2020) Opportunistic osteoporosis screening using chest radiographs with deep learning: development and external validation with a cohort dataset. J Bone Miner Res 37(2):369–377

Ji S, Lee B, Yi MY (2021) Building life-span prediction for life cycle assessment and life cycle cost using machine learning: a big data approach. Build Environ 205:108267

Jia X, Li Y, Shen B, Zheng W (2022) Evaluation, fabrication and dynamic performance regulation of green EMI-shielding materials with low reflectivity: a review. Compos B Eng 233:109652

Jiang, X., Huang, J., Jin, S., & Lu, S. (2023). Domain generalization via balancing training difficulty and model capability. In: Proceedings of the IEEE/CVF international conference on computer vision (pp. 18993–19003).

Juan Y, Dai Y, Yang Y, Zhang J (2021) Accelerating materials discovery using machine learning. J Mater Sci Technol 79:178–190

Karmaker N, Maraz KM, Islam F, Haque MM, Razzak M, Mollah MZI, Faruque MRI, Khan RA (2021) Fundamental characteristics and application of radiation. GSC Adv Res Rev 7(1):064–072

Kaşkaş A, Şahmaran T (2022) Determination of the radiological properties of materials: A new approximation method for calculation of the mass attenuation coefficients. Appl Radiat Isot 187:110340

Kassab MM (2022) Effect of chromium contents on radiation shielding and macroscopic cross-section in steel alloys. Appl Radiat Isot 186:110263

Kawamura M, Kamomae T, Yanagawa M, Kamagata K, Fujita S, Ueda D, Matsui Y, Fushimi Y, Fujioka T, Nozaki T, Yamada A (2024) Revolutionizing radiation therapy: the role of AI in clinical practice. J Radiat Res 65(1):1–9

Khosravi H, Sani KG, Jafari S, Nikzad S (2023) Evaluation of the radiation protection capability of a low density and non-Lead composite shield using Monte Carlo model. Int J of Radiation Res 21(4):833–836

Killeen P, Kiringa I, Yeap T, Branco P (2024) Corn grain yield prediction using UAV-based high spatiotemporal resolution imagery, machine learning, and spatial cross-validation. Remote Sensing 16(4):683

Kim SC (2021) Analysis of shielding performance of radiation-shielding materials according to particle size and clustering effects. Appl Sci 11(9):4010

Kocyigit F, Kaya AM (2023) Thermal and mechanical properties of a new insulation composite material. Materials Testing 65(9):1453–1463

Krichen M (2023) Convolutional Neural Networks 12(8):151

Laanen, P., Cuypers, A., Saenen, E., & Horemans, N. (2023). Flowering under enhanced ionising radiation conditions and its regulation through epigenetic mechanisms. *Plant Physiology and Biochemistry*.

Lakshminarayana G, Kumar A, Tekin HO, Issa SA, Al-Buriahi MS, Lee DE, Yoon J, Park T (2020) Binary B2O3–Bi2O3 glasses: scrutinization of directly and indirectly ionizing radiations shielding abilities. J Market Res 9(6):14549–14567

Lauwers, I., Pachler, K.S., Capala, M.E., Sijtsema, N.D., Van Gent, D.C., Rovituso, M., Hoogeman, M.S., Verduijn, G.M. and Petit, S.F., 2024. Ex vivo radiation sensitivity assessment for individual head and neck cancer patients using deep learning-based automated nuclei and DNA damage foci detection. *Clinical and Translational Radiation Oncology*, p.100735.

Lebedev, V., Shestopalov, O., Vyrovets, S., Masikevych, A., & Baranova, A. (2023, June). Electromagnetic Radiation Absorption Polymer Ceramic-Inorganic Composites Mechanical Properties Optimization. In *International Conference on Smart Technologies in Urban Engineering* (pp. 403–412). Cham: Springer Nature Switzerland.

Lee S, Lee S, Choi G, Lee B (2021) Analysis of Radiation Shielding Effect of Soft Magnetic Material applied to Military Facility. Journal of the Korean Society of Radiology 15(2):191–199

Lee, S.H., Geng, H., Arnold, J., Caruana, R., Fan, Y., Rosen, M.A., Apte, A.P., Deasy, J.O., Bradley, J.D. and Xiao, Y., 2023. Interpretable Machine Learning for Choosing Radiation Dose-volume Constraints on Cardio-pulmonary Substructures Associated with Overall Survival in NRG Oncology RTOG 0617. *International Journal of Radiation Oncology* Biology* Physics*.

Leyva JA, Munévar E (2023) Current status and development of neutron radiation for biophysical applications in Colombia. Biophys Rev 15(4):531–538

Li S, Liu L, Jiang Y, Tang C, Gu C, Li Z (2022) Ultrathin optically transparent electromagnetic shielding window with broadband microwave absorption and ultrahigh optical transmittance. Int J RF Microwave Comput Aided Eng 32(11):e23338

Li W, Nguyen-Thanh N, Du H, Zhou K (2023) Adaptive phase-field modeling of dynamic brittle fracture in composite materials. Compos Struct 306:116589

Lin YH, Okada M, Seifnashri S, Tachikawa Y (2023) Asymptotic density of states in 2d CFTs with non-invertible symmetries. J High Energy Phys 2023(3):1–43

Liu B, Lv H, Li L, Chai X, Xu H, Tan Y, Xia B, Chen C (2022a) Multi-objective optimization design of radiation shadow shield for space nuclear power with genetic algorithm. Frontiers in Energy Research 10:800930

Liu C, Benotto M, Ungar K, Chen J (2022b) Environmental monitoring and external exposure to natural radiation in Canada. J Environ Radioact 243:106811

Lu L, Sun M, Lu Q, Wu T, Huang B (2021) High energy X-ray radiation sensitive scintillating materials for medical imaging, cancer diagnosis and therapy. Nano Energy 79:105437

Lu Y, Zhang R, Wang L, Su X, Zhang M, Li H, Li S, Zhou J (2023) Prediction of diffuse solar radiation by integrating radiative transfer model and machine-learning techniques. Sci Total Environ 859:160269

Luoni F, Boscolo D, Fiore G, Bocchini L, Horst F, Reidel CA, Schuy C, Cipriani C, Binello A, Baricco M, Giraudo M (2022) Dose attenuation in innovative shielding materials for radiation protection in space: measurements and simulations. Radiat Res 198(2):107–119

Malekzadeh R, Sadeghi Zali V, Jahanbakhsh O, Okutan M, Mesbahi A (2020) The preparation and characterization of silicon-based composites doped with BaSO4, WO3, and PbO nanoparticles for shielding applications in PET and nuclear medicine facilities. Nanomedicine Journal 7(4):324–334

Malidarre RB, Akkurt I, Malidarreh PB, Arslankaya S (2022) Investigation and ANN-based prediction of the radiation shielding, structural and mechanical properties of the Hydroxyapatite (HAP) bio-composite as artificial bone. Radiat Phys Chem 197:110208

Maqbool, M. (Ed.). (2023). *An Introduction to Non-Ionizing Radiation*. Bentham Science Publishers.

Mathew, J., Kshirsagar, R., Abidin, D. Z., Griffin, J., Kanarachos, S., James, J., ... & Fitzpatrick, M. E. (2023). A comparison of machine learning methods to classify radioactive elements using prompt-gamma-ray neutron activation data. *Scientific Reports*, *13*(1), 9948.

Matsuya Y, Saga R (2023) Biophysical simulations for estimating biological effects after exposure to ionizing radiation: current state and future prospects. Radiation Environ Med 12(2):81–90

Mattos GRS, Bordon CDS, Kassab LRP, Issa SA, ALMisned G, Tekin HO (2023) Towards obtaining the optimum physical, optical and nuclear radiation attenuation behaviours of tellurite–germanate glasses through Eu2O3 reinforcement: glass synthesis, experimental and theoretical characterisation study. Ceramics International, 49(1), 986-99

Melia, E., & Parsons, J. L. (2023). DNA damage and repair dependencies of ionising radiation modalities. *Bioscience Reports*, *43*(10), BSR20222586.

Merchant, A., Batzner, S., Schoenholz, S. S., Aykol, M., Cheon, G., & Cubuk, E. D. (2023). Scaling deep learning for materials discovery. *Nature*, 1–6.

Morales, M. A., Henry, T. C., & Salamanca-Riba, L. G. (2023). Model of electromagnetic interference shielding effectiveness for a multifunctional composite containing carbon-fiber-reinforced polymer and copper mesh layers. *Carbon*, 118179.

More CV, Alsayed Z, Badawi MS, Thabet AA, Pawar PP (2021) Polymeric composite materials for radiation shielding: a review. Environ Chem Lett 19:2057–2090

Morgan D, Pilania G, Couet A, Uberuaga BP, Sun C, Li J (2022) Machine learning in nuclear materials research. Curr Opin Solid State Mater Sci 26(2):100975

Mostafa AMA, Issa SA, Zakaly HM, Zaid MHM, Tekin HO, Matori KA, Sidek HAA, Elsaman R (2020) The influence of heavy elements on the ionizing radiation shielding efficiency and elastic properties of some tellurite glasses: theoretical investigation. Results Phys 19:103496

Naito M, Kitamura H, Koike M, Kusano H, Kusumoto T, Uchihori Y, Endo T, Hagiwara Y, Kiyono N, Kodama H, Matsuo S (2021)

Applicability of composite materials for space radiation shielding of spacecraft. Life Sci Space Res 31:71–79
Naito, M., Kodaira, S., Ogawara, R., Tobita, K., Someya, Y., Kusumoto, T., ... & Orimo, S. I. (2020). Investigation of shielding material properties for effective space radiation protection. *Life sciences in space research*, *26*, 69–76.
Narayanan S, Zhang Y, Aslani F (2023) Prediction models of shielding effectiveness of carbon fibre reinforced cement-based composites against electromagnetic interference. Sensors 23(4):2084
Nath, K., Bhattacharyya, S. K., & Das, N. C. (2020). Biodegradable polymeric materials for EMI shielding. In *Materials for Potential EMI Shielding Applications* (pp. 165–178). Elsevier.
Nazar S, Yang J, Amin MN, Khan K, Ashraf M, Aslam F, Javed MF, Eldin SM (2023) Machine learning interpretable-prediction models to evaluate the slump and strength of fly ash-based geopolymer. J Market Res 24:100–124
Novello P, Poëtte G, Lugato D, Peluchon S, Congedo PM (2024) Accelerating hypersonic reentry simulations using deep learning-based hybridization (with guarantees). J Comput Phys 498:112700
Nyakyi CP, Mpeshe SC, Dida MA (2024) Assessment of public awareness on the effects of exposure to non-ionizing radiation sources in Tanzania. J Radiat Res Appl Sci 17(1):100770
Obrador E, Montoro A (2023) Ionizing radiation, antioxidant response and oxidative damage: radiomodulators. Antioxidants 12(6):1219
Oh S, Ahn JH, Jung R, Kim HJ, Chu Y, Choi DH, Lee H, Jung HD (2023) Characterization of various stainless steels containing gadolinium as thermal neutron absorbing and shielding materials. Metals 14(1):16
Okafor CE, Okonkwo UC, Okokpujie IP (2021) Trends in reinforced composite design for ionizing radiation shielding applications: a review. J Mater Sci 56:11631–11655
Okafor, C.E., Sunday, I., Ani, O.I., Ahmad, S., Mehfuz, S., Ekwueme, G.O., Chukwumuanya, O.E., Abonyi, S.E., Ekengwu, I.E. and Chikelu, O.P., 2023. Advances in machine learning-aided design of reinforced polymer composite and hybrid material systems. *Hybrid Advances*, p.100026.
Okonkwo UC, Ohagwu CC, Aronu ME, Okafor CE, Idumah CI, Okokpujie IP, Chukwu NN, Chukwunyelu CE (2022) Ionizing radiation protection and the linear No-threshold controversy: extent of support or counter to the prevailing paradigm. J Environ Radioact 253:106984
Oksengendler BL, Ashirmetov AK, Iskandarova FA, Zatsepin AF, Nikiforova NN, Suleimanov SK, Turaeva NN (2023) Interaction of radiation with hierarchical structures. J Surf Invest 17(1):31–42
Omer H (2021) Radiobiological effects and medical applications of non-ionizing radiation. Saudi J Biol Sci 28(10):5585–5592
Özcan, M., Avcıoğlu, S., Kaya, C., & Kaya, F. (2023). Boron carbide reinforced electrospun nanocomposite fiber mats for radiation shielding. *Polymer Composites*.
Özdemir HG, Kaçal MR, Akman F, Polat H, Agar O (2023) Investigation of gamma radiation shielding characteristics of bismuth reinforced ternary composites in wide photon energy region. Radiat Phys Chem 208:110924
Park G, Kim S, Park GK, Lee N (2021) Influence of carbon fiber on the electromagnetic shielding effectiveness of high-performance fiber-reinforced cementitious composites. J Build Eng 35:101982
Pathak, A. (2023). Interaction of Radiation with Matter. In *Tools and Techniques in Radiation Biophysics* (pp. 75–92). Singapore: Springer Nature Singapore.
Pathan MS, Pradhan SM, Selvam TP, Sapra BK (2024) A multistage machine learning algorithm for estimating personal dose equivalent using thermoluminescent dosimeter. Mach Learn 5(1):015011
Pradhan NC, Koziel S (2024) Shielded HMSIW-based frequency-tunable self-quadruplexing antenna using different solid/liquid dielectrics. AEU-Int J Electron Commun 175:155071
Qi S, Zhao W, Chen Y, Chen W, Li J, Zhao H, Xiao W, Ai X, Zhang K, Wang S (2022) Comparison of machine learning approaches for radioisotope identification using NaI (TI) gamma-ray spectrum. Appl Radiat Isot 186:110212
Qiao Y, Fring LD, Pallaka MR, Simmons KL (2023) A review of the fabrication methods and mechanical behavior of continuous thermoplastic polymer fiber–thermoplastic polymer matrix composites. Polym Compos 44(2):694–733
Riva S, Introini C, Cammi A (2024) Multi-physics model bias correction with data-driven reduced order modelling techniques: application to nuclear case studies. arXiv preprint arXiv:2401.07300.
Rudychev VG, Azarenkov MO, Girka IO, Rudychev YV (2021) The efficiency of radiation shielding made from materials with high atomic number and low mass density. Problems of Atomic Science and Technology. Series "Nucl. Phys. Invest, (2), 132.
Ruprecht NA, Singhal S, Schaefer K, Panda O, Sens D, Singhal SK (2024) A review: multi-omics approach to studying the association between ionizing radiation effects on biological aging. Biology 13(2):98
Sadhu, B., Sadhu, T., & Anand, S. (2024). RadDQN: a Deep Q Learning-based Architecture for Finding Time-efficient Minimum Radiation Exposure Pathway. *arXiv preprint* arXiv:2402.00468.
Şakar E, Alim B, Özpolat ÖF, Şakar BC, Baltakesmez A, Akbaba U (2021) A surveying of photon and particle radiation interaction characteristics of some perovskite materials. Radiat Phys Chem 189:109719
Salehi AW, Khan S, Gupta G, Alabduallah BI, Almjally A, Alsolai H, Siddiqui T, Mellit A (2023) A study of CNN and transfer learning in medical imaging: advantages, challenges. Future Scope Sustainability 15(7):5930
Saraswat, V., Dahshan, A., Elsaeedy, H. I., Khattari, Z., & Mehta, N. (2024). High-energy radiation shielding characteristics of SeTeSnM (M= Ag, Bi, Cd, Zn) chalcogenide glasses (STSM ChGs). *Ceramics International*.
Sayyed MI (2024) Effect of WO_3 on the attenuation parameters of TeO2–La2O3-WO3 glasses for radiation shielding application. Radiat Phys Chem 215:111319
Sayyed MI, Benhadjira A, Bentouila O, Aiadi KE (2024) Predicting the effective atomic number of glass systems using machine learning algorithms. Radiat Phys Chem 217:111479
Sayyed, M. I. (2023). The impact of Chemical composition, density and thickness on the Radiation Shielding properties of CaO–Al2O3–SiO2 glasses. *Silicon*, 1–10.
Schoenholz C, Zobeiry N (2024) An accelerated process optimization method to minimize deformations in composites using theory-guided probabilistic machine learning. Compos A Appl Sci Manuf 176:107842
Schütz, M., Schütz, A., Bendix, J., & Thies, B. (2023). Improving classification-based nowcasting of radiation fog with machine learning based on filtered and preprocessed temporal data. *Quarterly Journal of the Royal Meteorological Society*.
Scott H, Gallagher S, Abbott W, Talboys M (2022) Assessment of occupational dose reduction with the use of a floor mounted mobile lead radiation protection shield. J Radiol Prot 42(3):033501
Shahzad K, Kausar A, Manzoor S, Rakha SA, Uzair A, Sajid M, Arif A, Khan AF, Diallo A, Ahmad I (2022) Views on radiation shielding efficiency of polymeric composites/nanocomposites and multi-layered materials: current state and advancements. Radiation 3(1):1–20
Sharma A, Mukhopadhyay T, Rangappa SM, Siengchin S, Kushvaha V (2022a) Advances in computational intelligence of polymer composite materials: machine learning assisted modeling, analysis and design. Arch Comput Methods Eng 29(5):3341–3385

Sharma V, Zaki M, Jha KN, Krishnan NA (2022b) Machine learning-aided cost prediction and optimization in construction operations. Eng Constr Archit Manag 29(3):1241–1257

Shen, J., & Zhu, Q. (2022). Stability strategies of perovskite quantum dots and their extended applications in extreme environment: A review. *Materials Research Bulletin*, 111987.

Shi M, Feng CP, Li J, Guo SY (2022) Machine learning to optimize nanocomposite materials for electromagnetic interference shielding. Compos Sci Technol 223:109414

Shi K, Touge Y, Dou Y (2023a) Wildfire univariate and bivariate characteristics simulation based on multiple machine learning models and applicability analysis of wildfire models. Progress Disaster Sci 20:100301

Shi M, Peng W, Zhang Y, Tu Y, Zhang J, Li J, Guo S (2023b) Two-layer structural method: An efficient strategy inspired by machine learning for nanocomposites with magnetic particles for electromagnetic interference shielding. Compos Sci Technol 240:110082

Shuryak I, Nemzow L, Bacon BA, Taveras M, Wu X, Deoli N, Ponnaiya B, Garty G, Brenner DJ, Turner HC (2023) Machine learning approach for quantitative biodosimetry of partial-body or total-body radiation exposures by combining radiation-responsive biomarkers. Sci Rep 13(1):949

Siddique S, Ruda HE, Chow JC (2023) FLASH radiotherapy and the use of radiation dosimeters. Cancers 15(15):3883

Sidi-Salah L, Chouai M, Danlée Y, Huynen I, Ouslimani N (2020) Simulation and optimization of electromagnetic absorption of polycarbonate/CNT composites using machine learning. Micromachines 11(8):778

Sobczak, J., & Żyła, G. (2024). Nano and microcomposites as gamma and X-ray ionizing radiation shielding materials—A review. *Energy*, 130210.

Sreenivas, B., kareem Ahmmad, S., Rammah, Y. S., & Bindu, P. H. (2023). Radiation shielding capacity and machine learning density prediction of boro-bismuth cadmium zinc glasses. *Open Ceramics*, *16*, 100493

Sterratt D, Graham B, Gillies A, Einevoll G, Willshaw D (2023) Principles of computational modelling in neuroscience. Cambridge University Press

Sun W, Hu G, Xu H, Li Y, Wang C, Men T, Ji F, Lao W, Yu B, Sheng L, Li J (2022) Study on the influence of reinforced particles spatial arrangement on the neutron shielding performance of the composites. Materials 15(12):4266

Sun L, Li L, Fu Q, Qian X (2023) Green fabrication of transparent Copper (II)-imidazolate metal–organic framework/cellulose nanocomposite films for antibacterial and UV-shielding applications. Cellulose 30(6):3569–3588

Tamam N, Alrowaili ZA, Olarinoye IO, Hammoud A, Albarkaty KS, Kebaili I, Al-Buriahi MS (2024) Fabrication and characterisation of TeO2-based composite doped with Yb3+ and Bi3+ for enhanced radiation shielding safety. Radiat Phys Chem 215:111315

Tan X, Zhang B, Liu K, Yan X, Han J, Liu X, Yang W, Zhou C, Yu Z, Shao P, Wu G (2020) Microstructure and mechanical property of the 2024Al matrix hybrid composite reinforced with recycled SiCp/2024Al composite particles. J Alloy Compd 815:152330

Tougui I, Jilbab A, El Mhamdi J (2021) Impact of the choice of cross-validation techniques on the results of machine learning-based diagnostic applications. Healthcare Inform Res 27(3):189–199

Tsaples G, Papathanasiou J, Georgiou AC (2022) An exploratory DEA and machine learning framework for the evaluation and analysis of sustainability composite indicators in the EU. Mathematics 10(13):2277

Turczyn R, Krukiewicz K, Katunin A, Sroka J, Sul P (2020) Fabrication and application of electrically conducting composites for electromagnetic interference shielding of remotely piloted aircraft systems. Compos Struct 232:111498

Valizadeh, A., Amirhosseini, M. H., & Ghorbani, Y. (2024). Predictive Precision in Battery Recycling: Unveiling Lithium Battery Recycling Potential through Machine Learning. *Computers & Chemical Engineering*, 108623.

Varone, G., Ieracitano, C., Çiftçioğlu, A.Ö., Hussain, T., Gogate, M., Dashtipour, K., Al-Tamimi, B.N., Almoanmari, H., Akkurt, I. and Hussain, A., 2023. A novel hierarchical extreme machine-learning-based approach for linear attenuation coefficient forecasting. *Entropy*, *25*(2).

Vinayagamoorthy, R. (2022). Bio-fibre Reinforced Polymeric Composites for Industrial, Medicine and Domestic Applications. *Bio-Fiber Reinforced Composite Materials: Mechanical, Thermal and Tribological Properties*, 31–49.

Wang M, Tang XH, Cai JH, Wu H, Shen JB, Guo SY (2021) Construction, mechanism and prospective of conductive polymer composites with multiple interfaces for electromagnetic interference shielding: a review. Carbon 177:377–402

Wang L, Zhang X, Li Q, Zhang M, Su H, Zhu J, Zhong Y (2023) Incorporating neuro-inspired adaptability for continual learning in artificial intelligence. Nat Mach Intell 5(12):1356–1368

Wang L, Pan Q, Wang S (2024) Data-driven predictions of shield attitudes using Bayesian machine learning. Comput Geotech 166:106002

Willard J, Jia X, Xu S, Steinbach M, Kumar V (2020) Integrating physics-based modeling with machine learning: a survey. arXiv preprint arXiv:2003.04919, 1(1), 1–34.

Wu Y, Liu B, Su X, Tang S, Yan M, Pan L (2024) Study on multi-objective optimization method for radiation shield design of nuclear reactors. Nucl Eng Technol 56(2):520–525

Xu J, Zhang Z, Zhang L, Liu D (2023) Predicting shield position deviation based on double-path hybrid deep neural networks. Autom Constr 148:104775

Yang DY, Frangopol DM, Han X (2021a) Error analysis for approximate structural life-cycle reliability and risk using machine learning methods. Struct Saf 89:102033

Yang F, Tang ZR, Chen J, Tang M, Wang S, Qi W, Yao C, Yu Y, Guo Y, Yu Z (2021b) Pneumoconiosis computer aided diagnosis system based on X-rays and deep learning. BMC Med Imaging 21:1–7

Yang H, Fang L, Yuan Z, Teng X, Qin H, He Z, Wan Y, Wu X, Zhang Y, Guan L, Meng C (2023) Machine learning guided 3D printing of carbon microlattices with customized performance for supercapacitive energy storage. Carbon 201:408–414

Yılmaz D, Kurucu Y, Gürol A (2023) Effective atomic numbers and number, energy and dose albedos of different colored thermoplastic materials. Radiat Eff Defects Solids 178(3–4):325–334

Yuhang Z, Yu Z, He H, Ke H (2024) Multifunctional electromagnetic interference shielding composite fabrics with simultaneous high-efficiency photothermal conversion and Joule heating performance. Pigm Resin Technol 53(2):192–200

Yunus NAM, Sazali MA, Yazid K, Salleh WNW, Abdullah MZ, Idris FM, Ali NSM, Sarkawi MS, Jamaluddin K (2023) Analysis of boron epoxy resin composite in morphology, neutron radiography, and thermal neutron shielding performance. Radiat Phys Chem 204:110670

Zhang C, Hua L, Ji C, Nazir MS, Peng T (2022a) An evolutionary robust solar radiation prediction model based on WT-CEEMDAN and IASO-optimized outlier robust extreme learning machine. Appl Energy 322:119518

Zhang P, Tian R, Zhang X, Ding X, Wang Y, Xiao C, Zheng K, Liu X, Chen L, Tian X (2022b) Electromagnetic interference shielding epoxy composites with satisfactory thermal conductivity and electrical insulation performance enabled by low-melting-point alloy layered structure. Compos B Eng 232:109611

Zheng SC, Pan QQ, Lv HW, Tang SQ, Liu XJ (2023) Semi-empirical and semi-quantitative lightweight shielding design method. Nucl Sci Tech 34(3):43

Zhu X, Wan Z, Tsang DC, He M, Hou D, Su Z, Shang J (2021) Machine learning for the selection of carbon-based materials for tetracycline and sulfamethoxazole adsorption. Chem Eng J 406:126782

Zhu Z, Wang J, Jiao Y, Shao C, He D, Hu L (2023) Effects of the ionizing irradiation of γ-rays, X-rays, and UV-rays on the optical properties of Ag-contained/Ag-free photo-thermo-refractive glass. Ceram Int 49(2):1826–1835

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